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Investigation of the floating IEA Wind 15 MW RWT using vortex methods Part I: Flow regimes and wake recovery

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Abstract

Floating wind turbines have the potential to enable global exploitation of offshore wind energy, but there is a need to further understand the complex aerodynamic phenomena they can encounter due to floater induced rotor motion. Aerodynamic models traditionally used in the wind energy sector, like the Blade Element Momentum (BEM) theory, may not be capable of capturing the dynamic phenomena that occur when the rotor moves in and out of its own wake. In the present paper, we therefore compare an industry standard BEM-based code to a state of the art vortex solver, to investigate the phenomena in detail and further clarify the capabilities and limitations of both methods. An initial benchmark of the two codes using the IEA Wind 15 MW RWT mounted on the WindCrete spar-buoy floater is carried out. Three different scenarios are taken into account: a bottom-fixed, a floating case, and a floating case subject to regular waves. Growing discrepancies between the codes have been observed with the increasing complexity of the simulations. Moreover, large differences between the wake generated by a bottom fixed and a floating turbine have been observed, with the latter one experiencing a faster recovery. To further explore the floating turbines behaviors that can affect the rotor performance and wake, a systematic investigation of the mean tilt angle influence in the wake development has been carried out. Further, to account for the oscillatory motion of a floating turbine, a parametric study where the floater motion is prescribed in both pitch and surge degrees of freedom (DoF) is designed. The study covers a large variety of scenarios; a wide range of relevant frequencies and amplitudes are taken into account in under and above-rated wind conditions. A total of more than 28 unique cases have been defined and simulated with both fidelity models. The results include the downstream evolution of the wake recovery and intensity of the turbulent disturbances induced by the rotor. The generic nature of the study allows to characterize the flow and performance effects and enables subsequent generalization to floater designs of given natural frequency and motion amplitude. It has been found that the BEM and LL predictions of the maximum loading in the blade root and tower bottom compare quite well, except for the case of large oscillation frequency in above rated

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conditions, where the BEM method under-predicts the loads. Moreover, the use of a vortex solver makes it possible to look in depth into the wake characteristics, in which large differences are observed between the bottom-fixed and the floating case in non-turbulent inflow conditions. It has been found that the frequency and amplitude of the turbine oscillations can have a strong impact on the recovery of the wake. Moreover, we have found a link between short time intervals of large FA blade tip displacements and a faster break down of the wake. Finally it has been shown that a floating wind turbine with large and fast oscillations can transition between the different rotor states, including the non-conventional propeller and vortex ring states, which are identified and characterized.

KEYWORDS

BEM, floating wind turbine, lifting line, offshore wind, vortex method, wakes

1 | INTRODUCTION

The immense offshore wind resource has the potential to power the world, attracting more and more the attention of both industry and academia. However, the majority of the available offshore wind is located in deep waters, making floating turbines a key technology for the further exploitation of offshore wind energy. Moreover, in contrast with onshore wind, offshore wind generally provides higher wind speeds and lower turbulence levels in free atmospheric flow conditions.^{1,2} Furthermore, regulatory esthetic and noise requirements are much lower, making possible the installation of larger wind turbines.^{1,3,4} The floaters for offshore wind deployment are usually classified into spar-buoys, semi-submersibles, barges, and tension-leg platforms (TLPs). Whereas TLPs are very stiff with respect to floater pitch motion, the other floater types are typically restrained with a catenary mooring system, thus allowing large motions around an equilibrium point in both surge and pitch.

During the design and planning stages of new offshore wind turbines and wind farm development, simulation tools are used to predict the coupled dynamic loads and response of the offshore wind turbine structure.⁵ Such tools are developing very quickly in recent years and vortex solvers have the potential to become a key method due to their flexibility and reliability.^{6–8} With these solvers, it is possible to easily combine different flow and aerodynamic models depending on the degree of complexity required for the case of study. Enumerated in ascending complexity, the aerodynamic models traditionally implemented in the vortex solvers can be classified as Lifting Line (LL), vortex lattice, potential panel, viscous-inviscid panel, and penalization methods. Analogously, one can divide the flow solvers used in prescribed filament wake methods, free wake filament methods, hybrid filament-particle-mesh methods, and particle-mesh methods. For the present investigation a LL model in combination with a hybrid filament-particle-mesh method is used.

Blade Element Momentum theory (BEM), which at the moment is considered as the industry standard, is not capable to accurately simulate complex blade geometries nor complex flow cases, e.g., yaw and shear. Sub-models to account for such complexities need to be implemented and calibrated with higher fidelity codes or even experimental data. For example, it is not yet clear that the current dynamic inflow models are valid under the most severe unsteady effects a floating machine is subject to. Vaal et al⁹ found that dynamic inflow models calibrated for rapidly changing loads were suitable to be used for analyzing surging wind turbines subject to the low frequency oscillations representative of a TLP. It was found that differences between a BEM and an actuator disc model increased with the increasing oscillation frequency. Other studies like the one of Sivalingam et al¹⁰ have observed larger differences of the BEM based predictions compared to unsteady CFD, in line with the observations of Tran and Kim.¹¹ Moreover, previous work of Sebastian and Lackner¹² on floating machines found that slip-stream violations are most likely to occur for the outboard blade sections at low wind speeds or high tip-speed ratios, demonstrating the necessity of higher-fidelity tools. Moreover, the large fluctuations in the local relative velocities seen by the oscillating blades make floating turbines more likely to encounter “off-design” rotor states, i.e., turbulent wake, vortex ring, and propeller states. Note that some of these states have been investigated in detail using high fidelity CFD solvers in Kyle et al¹³ and Lienard et al.¹⁴ These states present a challenge for standard BEM methods since they can violate the momentum balance assumption¹⁵ due to the appearance of local re-circulation areas or due to the large expansion of the wake at high tip speed ratios, which causes the flow field behind the rotor to be governed by turbulent mixing. Despite these challenges, BEM-based methods can be used to predict the onset of vortex ring state in helicopters¹⁶ and wind turbine applications.¹⁷ It is presumed that empirical corrections have to be used to correctly model these flows in BEM. Sørensen et al¹⁸ compared one-dimensional momentum theory against a Navier-Stokes algorithm with an actuator disc method to model the rotor, finding with their simulations that the turbulent wake and vortex ring states were unstable regimes with a very complicated transient phase.

Kopperstad et al¹⁹ studied the wakes of floating wind turbines using an actuator disc LES methodology. They investigated two concepts, a spar buoy and a barge platform, finding that the motion of the barge platform induced a faster wake recovery in both laminar and turbulent inflow conditions. Rockel et al^{20,21} conducted wind tunnel experiments with stereo Particle Image Velocimetry (PIV) to analyze the effect of pitch motion on the wake development of horizontal axis wind turbines. Their results indicate an upward shift of the wake as a result of the turbine being tilted upwards that is consistent with this study. However, contrary to our findings, they report that the wake of the pitching turbine is governed by an increase in the mean vertical component of velocity that decreases the turbulence production, suppresses the entrainment of kinetic energy, and reduces the mean streamwise velocity. Fu et al²² performed laboratory experiments to study the impact of pitch and roll motion on the power output and wake of a horizontal axis wind turbine. Their results, as ours, show a faster wake recovery for the oscillating turbine due to an enhanced turbulent mixing that leads to an increased turbulence kinetic energy.

Floating wind turbines are subject to large periodic oscillations, which can trigger such complex transient phenomena. It is therefore necessary to employ higher-fidelity tools capable to simulate such complex cases and use the obtained knowledge to improve and calibrate the faster engineering methods. Although there is a strong commercial interest in the design of floating wind farms, the use of simplified engineering tools to predict the aerodynamic performance in the more dynamic conditions encountered in oscillatory rotor motion has not yet been verified in detail. Simple methods may miss the required accuracy to model such complex scenarios. The same also applies to wake modeling, where existing engineering methods do not include the effect of the floater motion. This is the purpose of the present study, where the in-house code MIRAS-HAWC2 has been used to perform fully interactive aero-hydro-servo-elastic simulations of the IEA Wind 15 MW Reference Wind Turbine (RWT)²³ mounted on the WindCrete spar-buoy floater²⁴ for different waves and wind conditions. Moreover, the effect of mean tilt in the development of the wake has been investigated, and a detailed parametric study on the rotor performance and the wake properties in prescribed harmonic motion has been carried out. This generic approach allows for characterization of the flows and generalization for various floater types based on natural frequencies and representative motion amplitudes. The study involved more than 28 different cases, including different combinations of wind speed and floater motion. Only the surge translation and the pitch rotation of the floater have been investigated in detail, since these two degrees of freedom are the most relevant in the response of a floating wind turbine to wind loading. For all simulated cases presented in this manuscript, the MIRAS-HAWC2 predictions are compared to those obtained with the BEM method implemented in HAWC2.

The manuscript is divided in four main sections, excluding the introduction. First, the numerical tools used throughout the investigation are presented in the methodology section. Second, the simulation setup is introduced including a definition of the IEA Wind 15 MW RWT, the WindCrete floater, the DTU controller calibration, the test matrix definition, and the computational setup. The results are presented in four independent sections. First, the simulations of the turbine mounted on the WindCrete floater are presented and analyzed. Second, the investigation of the effect of tilt in the wake development is presented. Finally, the parametric study using a prescribed harmonic motion of the floater is presented for pitch and surge degrees of freedom. Conclusions of the study are drawn in a final section.

2 | METHODOLOGY

In this section, the numerical methods involved in the simulations of the present work are introduced. First, details of the vortex solver MIRAS are given in Section 2.1. The integration of the aerodynamic and wake models of MIRAS with the multi-body finite-element solver HAWC2 is then described in Section 2.2. This allowed to account for the rest of the main physics involved in floating wind turbines, namely, the structural dynamics, the actuation of the wind turbine controller, and the effects of the hydrodynamics and the mooring system.

2.1 | MIRAS vortex solver

The in-house vortex code MIRAS,²⁵⁻²⁸ i.e., Method for Interactive Rotor Aerodynamic Simulations, has been the main solver employed to model the rotor aerodynamics and the flow. MIRAS is a multi-fidelity aerodynamic tool for wind turbine and wake analyses. In the herein presented research, the LL module is used as aerodynamic model, while the hybrid filament-particle-mesh method is used to model the flow.

In the LL method, the rotor blades are modeled as discrete filaments, which account for the bound vortex strength and act by releasing vorticity into the flow. This vorticity can be subdivided into trailing vorticity, which is related to span-wise variations of the bound vorticity and shed vorticity, related to time variations of the bound vortex. The bound vortex strength, Γ , is calculated applying the Kutta-Joukowski theorem at the different span-wise stations along the blade as follows:

$$\Gamma = \frac{L}{\rho V_{qc}}, \quad (1)$$

where $\underline{L}$ is the lift force of each aerodynamic section, obtained by interpolation in a set of tabulated airfoil data (C_l, C_d, C_m) as function of the computed angle of attack. Such two-dimensional (or 3D-corrected) airfoil data are the same as the one used in BEM models, making the comparison between such different fidelity tools very straight forward. ρ is the air density at a given temperature, and V_{qc} is the total velocity at the quarter chord of the different blade sections, calculated as follows:

$$\underline{V}_{qc} = \underline{V}_0 + \underline{V}_b + \underline{u}_w + \Delta \underline{u}^b, \quad (2)$$

where $\underline{V}_0$ is the inflow velocity, $\underline{V}_b$ is the blade motion, $\underline{u}_w$ is the wake induced velocity, and $\Delta \underline{u}^b$ accounts for a bound vortex correction applied to curved blade geometries and detailed in Ramos-García et al.²⁹

Øye's dynamic stall model³⁰ is used to account for the delay in the stall mechanisms due to dynamic inflow changes seen by the airfoils. Moreover, an additional non-circulatory contribution to the computed C_l and C_m ³¹ is included to account for the correct unsteady lift and blade pitch moment in dynamic attached flow. These corrections depend only on the instantaneous acceleration and velocity of the airfoil motion and account for the missing torsional damping incurred when computing the airfoil characteristics at the quarter chord.

$$C_l = C_{l0} + \frac{0.50 c \pi \dot{\alpha}}{V_{qc}}, \quad C_m = C_{m0} + \frac{0.25 c \pi \dot{\alpha}}{V_{qc}}, \quad (3)$$

where $\dot{\alpha}$ is the torsional rate. These non-circulatory terms have been found key throughout the investigation for the simulation of floating turbine in above-rated conditions. A violent torsional instability was observed when the corrections were neglected.

The initial sheet of vortex filaments is transformed into a set of vortex particles which vorticity is later on interpolated into an auxiliary Cartesian mesh. The downstream location of the filament to particle transformation depends on type of case we want to simulate. In this case, due to the aero-elastic nature of the problem and the necessity of small time steps, only one row of vortex filaments has been used. A study on the influence of this distance in a hybrid filament-mesh was previously published in Ramos-García et al.²⁷ The motion of the vortex elements is determined by the velocity of their markers $\underline{u}(\underline{x}_i)$ (filaments endpoints or the particles itself), which is calculated by a superposition of the free-stream velocity $\underline{V}_0$ and the velocity contributions from all vortex and boundary elements at the marker positions $\underline{x}_i$,

$$\frac{d\underline{x}_i}{dt} = \underline{u}(\underline{x}_i) \quad \text{with} \quad \underline{u}(\underline{x}_i) = \underline{V}_0 + \underline{u}_f(\underline{x}_i) + \underline{u}_{fw}(\underline{x}_i) + \underline{u}_{pw}(\underline{x}_i), \quad (4)$$

where $\underline{u}_f$ is the velocity induced by the blade bound vortex, $\underline{u}_{fw}$ is the velocity induced by the filament wake, and $\underline{u}_{pw}$ is the velocity induced by the particle wake, in which the Poisson equation is solved by a high-order regularization method introduced by Hejlesen et al.³² A detailed explanation of the different velocity contributions of Equation 4 and a study on the choice of the time-integration scheme was presented in Ramos-García et al.^{27,28}

The vorticity equation, curl of the Navier-Stokes equation, is used to account for stretching and diffusion of the vortex structures. It describes the evolution of the vorticity of a fluid particle as it moves with the flow, and assuming an incompressible fluid with a constant and uniform viscosity reads

$$\frac{D\underline{\omega}}{Dt} = \frac{\partial \underline{\omega}}{\partial t} + (\underline{u} \cdot \nabla) \underline{\omega} = (\underline{\omega} \cdot \nabla) \underline{u} + \nu \nabla^2 \underline{\omega}. \quad (5)$$

The left-hand side of the equation characterizes the rate of change of the vorticity, including the unsteady term, $\frac{\partial \underline{\omega}}{\partial t}$, and the convection term, $(\underline{u} \cdot \nabla) \underline{\omega}$, which accounts for the changes in vorticity due to the motion of the fluid particle. On the right-hand side, $(\underline{\omega} \cdot \nabla) \underline{u}$ accounts for the vortex stretching and $\nu \nabla^2 \underline{\omega}$ for the vortex diffusion due to viscous effects.

2.2 | Coupled simulations

In order to model the complex environment that a floating wind turbine is subjected to, the inclusion of different physics is required. Together with the rotor aerodynamics modeled with MIRAS (Section 2.1), the effects of the control actuation, the hydrodynamic loading, and the structural response were considered in the present work. This led to a coupled multi-physics problem, which is often referred to as aero-servo-hydro-elasticity in the literature. This problem was handled by the multi-body finite-element software HAWC2.³³ The framework that HAWC2 uses for the solution of the aero-servo-hydro-elasticity problem relies on an augmented form of the floating frame of reference (FFR) formulation (see for instance Shabana³⁴). In FFR, the kinematics of every flexible body is described in a coordinate system which translates and rotates relative to an

inertial system. In this way, each of the flexible components of the models developed within the present work was characterized by two complementary discretizations. On one hand, a set of linear isotropic Timoshenko beam elements was defined. On the other hand, an arbitrary number of bodies was also specified along the component. Those bodies were connected through a set of constraints that allowed to capture the geometrical non-linearities of the structural response under the presence of the aerodynamic loading, the hydrodynamic loading, and the control actuation. At every time step, HAWC2 solved the constrained equation of motion in order to obtain the corresponding generalized coordinates. This solution was performed iteratively, based on a predictor/corrector approach, resulting in a strong coupling between the different physics involved. A Newmark-beta scheme was used for the time-integration.³⁵

Figure 1 illustrates the integration of the MIRAS solver into the HAWC2 *aero-servo-elastic* solution, together with the rest of the physical models concerned by the present work. The aforementioned integration was done through an external Python code: the *DTU coupling*. This outer layer accessed the instances of MIRAS and HAWC2 codes, compiled as shared objects for this purpose. The *DTU coupling* orchestrated the execution of both solvers and managed the transfer of the wind turbine kinematics predicted by HAWC2 and the aerodynamic loads computed by MIRAS. MIRAS loads could then be injected into the built-in *aerodynamic model* of HAWC2, thus replacing the originally computed BEM loads. This allowed to re-use the already existing routines for loads interpolation and the aeroelastic solution of HAWC2, thus simplifying the implementation of the coupling. It should be remarked that the replacement of BEM blade loads by MIRAS loads occurred only once per time step, resulting in a loose aeroelastic coupling for this particular case. The reader is referred to Ramos-García²⁹ for more information regarding the aeroelastic coupling between MIRAS and HAWC2 and for a detailed comparison against in-house fully resolved FSI (Fluid Structure Interaction) simulations and other aeroelastic tools used in the framework of the OC3 and OC4 projects.^{36–38} This type of simulations are labeled in the present work as MIRAS-HAWC2, MIRAS, or simply LL.

Alternatively, the injection of the MIRAS blade loads could be switched off, so that pure BEM simulations could be made for comparison purposes. That type of computations are referred to in this document as HAWC2-BEM, HAWC2, or simply BEM. While BEM is considered a step-down in terms of modeling fidelity with regards to MIRAS, the implementation of the former approach in HAWC2 features several enhancements in order to increase its range of applicability. Those features include a modification to account for the wake expansion and swirl and extended models to handle: dynamic inflow, skewed inflow, shear effect on induction, the effect from large blade deflections, tip loss, and dynamic stall. In the present study, the dynamic stall model of Leishman and Beddoes³⁹ has been employed. The reader is referred to the recent publication of Madsen et al⁴⁰ for more details regarding the BEM implementation of HAWC2. Moreover, it has been found critical to include the effect of rotor motion for the yaw/tilt correction and the dynamic inflow time constants in BEM. It is important to note that during the course of this study, we

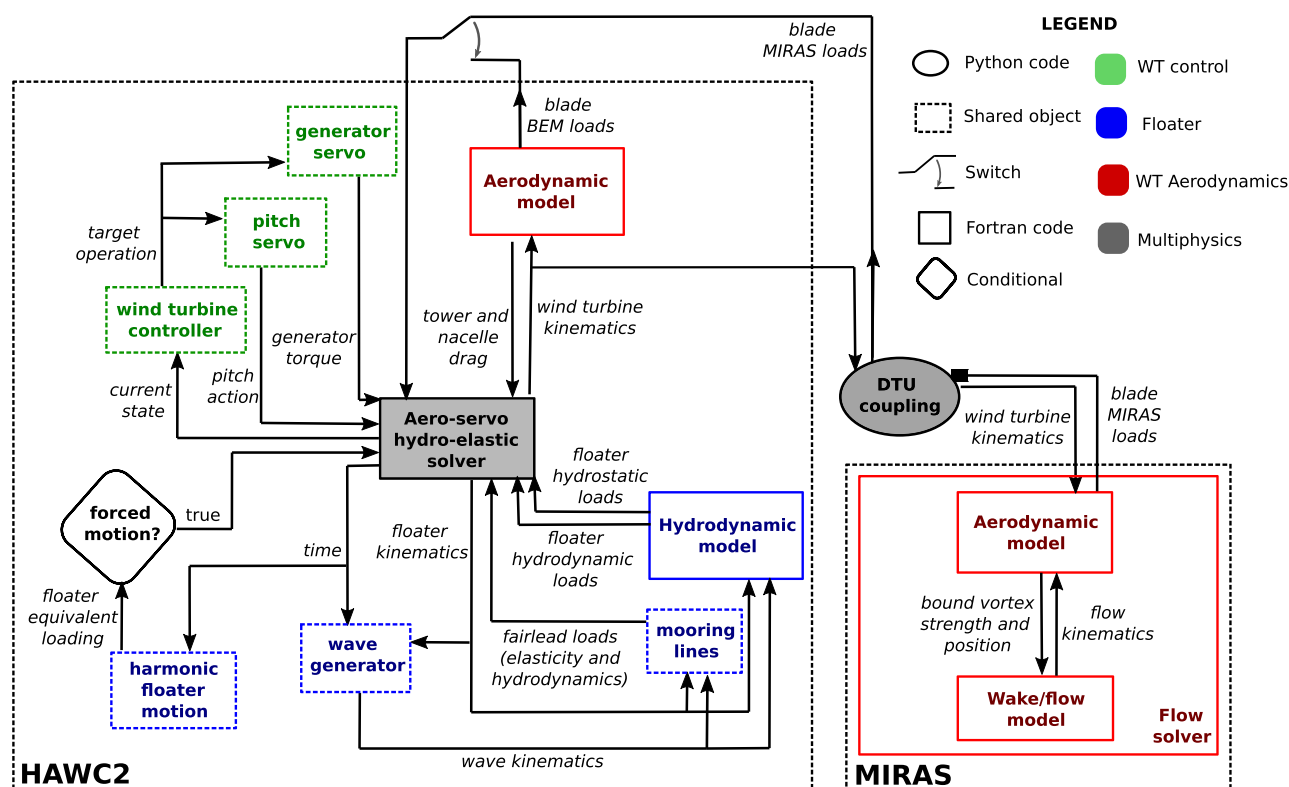


FIGURE 1 Sketch of the different components involved in the simulations

have seen large differences between MIRAS-HAWC2 and HAWC2-BEM when not using such corrections, especially at high frequencies of the pitch and surge floater motions.

The rest of the physical models involved in the coupled simulations were introduced making use of the standard capabilities of HAWC2. The built-in *hydrodynamic model*, relying on linear wave kinematics from a *wave generator* and the Morison equation,⁴¹ was used to account for the hydrodynamic loads on the floating substructure. This platform also received a set of fair-leads loads coming from a *mooring line model*. This model consisted on a series of elastic bars under the action of hydrodynamic loads and was coded as an external library. As described in Section 3.5, some configurations studied in the present work analyzed particular sea states or floating platform behaviors. Therefore, in some computations, the response of the moored floating wind turbine to wind and wave loads was calculated. In other simulations, the hydrodynamic and mooring modules were disabled, and the motion of the floating platform in different degrees of freedom was forced. This was achieved through the action of an external library, labeled here as *harmonic floater motion*, that computed a set of equivalent concentrated loads to achieve the desired forced motion. The wind turbine control was also handled by a stand alone shared object (*wind turbine controller*), corresponding to the basic implementation of the DTU Wind Energy Controller.⁴² The transformation of the controller output into a new collective blade pitch and a generator torque was steered by two additional libraries, labeled in this document as *pitch servo* and *generator servo*.

3 | CHOICE OF SIMULATION CASES

This section describes cases of study presented, including the IEA Wind 15 MW RWT turbine, the WindCrete spar-buoy floating platform, the DTU Wind Energy Controller (WEC) calibration, and the prescribed floater motion.

3.1 | IEA Wind 15 MW Reference Wind Turbine

The wind turbine chosen for this investigation is the recently released IEA Wind 15 MW RWT, especially designed for offshore applications and extensively described in Gaertner et al.²³ Some key figures are given in Table 1.

3.2 | WindCrete

WindCrete is a monolithic concrete spar-buoy that includes the tower and the floater in a continuous single piece. The advantage of such monolithic construction is the absence of joints between the tower and the floater, which leads to an increased fatigue resistance. The tower is a truncated cone with a constant thickness of 0.4 m and a height of 129.495 m resulting in a hub height of 135 m. A tapered transition piece connects the tower to a cylindrical spar with a diameter of 18.6 m and a length of 135.7 m. The bottom of the spar buoy is closed by a hemisphere of the same diameter. The total draft of the WindCrete is 155 m, and the total mass is $3.9805 \cdot 10^7$ kg including ballast. The main dimensions of the WindCrete are summarized in Figure 2, where the different DoF are also sketched. The station keeping system consists of three single crowfoot catenary mooring lines with 120° spacing. The rigid-body natural frequencies of the WindCrete floater with the IEA Wind 15 MW turbine are presented in Table 2, and more details about the floater can be found at Mahfouz et al.²⁴ and Vigara et al.⁴³

The structural modeling of WindCrete is based on the multibody formulation of HAWC2 where the structure is an assembly of bodies and each body is an assembly of Timoshenko beam elements. The hydrodynamic loads are based on the Morison equation that is suitable for slender

TABLE 1 Key figures for the IEA Wind 15 MW Reference Wind Turbine²³

Parameter	Value
Rated power	15 MW
Rated wind speed	10.59 m/s
Turbine class	IEC Class 1B
Rotor diameter	240 m
Design tip-speed ratio	9.0
Rotor speed range	5.00–7.56 rpm
3P excitation range	0.250–0.378 Hz
First blade flapwise natural frequency	0.555 Hz
First blade edgewise natural frequency	0.642 Hz

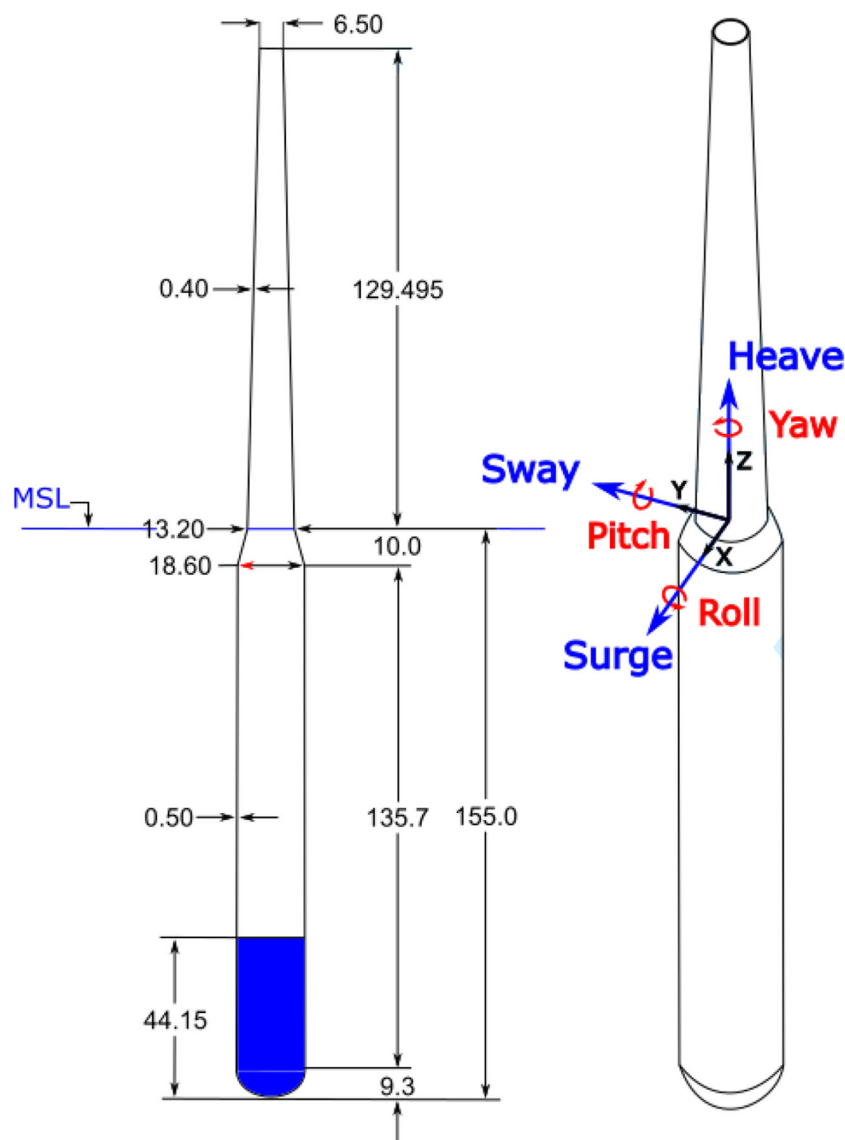


FIGURE 2 WindCrete Sketch with dimensions in meters²⁴

structures. The buoyancy loads calculation is based on the integration of the external water pressure on the submerged body. The mooring lines are modeled by a dynamic modeling approach using a general cable element formulation that includes hydrodynamic drag and buoyancy forces. In the paper, the herein presented floating wind turbine is compared against a fixed machine, in which the motion of the floater is eliminated. Practically, the floater has been clamped at the tower interface in the HAWC2 structural solver. This configuration is referred throughout the paper as a bottom-fixed turbine.

3.3 | DTU controller calibration

The HAWC2 model of WindCrete uses the DTU Wind Energy Controller.⁴² Using an onshore controller on a floating wind turbine can lead to blade pitch instability for above rated wind speeds. Such instability is directly related to the coupling of the tower motion and the blade pitch control.⁴⁴ Therefore, the controller was tuned using HAWCStab2 Version 2.16a.⁴⁵ The assumed partial-load poles (torque controller) were set to a natural frequency of 0.05 Hz, with 70% critical damping. The full-load poles (blade pitch controller) were set to 0.011 Hz natural frequency (half the floater's pitch frequency) and 70% critical damping. The control strategy combines constant power (90%) and constant torque (10 %); therefore, a linear interpolation between them is used to compute the generator torque limits. Such control strategy causes small variations in power in above rated wind conditions. The stability of the WindCrete with the new controller settings was first confirmed by doing a HAWC2 step-wind

TABLE 2 Rigid-body natural frequencies of the WindCrete floater, mounting the IEA Wind 15 MW wind turbine²⁴

Degree of freedom	Natural frequency [Hz]
Surge	0.01221
Heave	0.03052
Pitch	0.02441
Yaw	0.09155

test and simulations with steady wind above rated wind speed. The allowed max low-pass filtered tower top acceleration was increased in order to avoid shut down during the initial transient where the turbine undergoes large accelerations in the forced prescribed motion cases described in the following.

3.4 | Prescribed floater motion

The aerodynamic behavior of floating wind turbines is generally more complex than for bottom-fixed machines, mainly because the rotor experiences a significant motion due to the floating support structure and its interaction with the waves. From an aerodynamic point of view, the most relevant floater motions are those that result in an oscillation of the rotor in the direction of the wind. Assuming no misalignment, those motions correspond to the angular displacement of the platform referred to as *pitch* and to its translatory displacement in *surge*.

To investigate the effects of the floater motion on the rotor aerodynamics, parametric studies were carried out for pitch and surge. For each of these two DoFs, the displacement of the floater was prescribed as a sinusoidal function with user-defined amplitude and frequency. In the prescribed surge cases, the entire system translates back-and-forth along the X axis (see Figure 2), whereas in the prescribed pitch cases, it rotates around the point of flotation, located at the mean water level. We note that the instantaneous center of rotation of the structure is located below water, due to the very low center of mass of spar-buoy floaters. The choice of the present coordinate system, however, is common practice when modeling floating wind turbines.

The amplitude and frequencies of the two parametric studies have been chosen to cover a representative range of the pitch and surge motion of a floating wind turbine. In this way, the investigation can be generalized beyond the spar-buoy floater. For pitch (Table 3), the frequency study is carried out at 0.01419, 0.02838, 0.05676, and 0.11352 Hz with a pitch amplitude of 6.38°. The first two frequencies are representative of the surge and pitch natural frequency of floating platforms, whereas the last two frequencies represent wave peak frequencies in storm and normal conditions. Moreover, an amplitude study was performed for 3.19°, 6.38°, and 12.7° with a frequency of 0.02441 Hz, which is the pitch natural frequency of the WindCrete floater (see Table 2). Here, the first amplitude is representative of strong motion under turbine operation, while the two next are stronger cases, selected to study the amplitude effect beyond normal operation.

For surge (Table 4), the frequency study was carried out for the same frequencies as the pitch study and an amplitude of 15 m. The amplitude study includes surge amplitudes of 5, 10, and 15 m at a frequency of 0.01419 Hz, which is close to the surge natural frequency for the WindCrete platform (see Table 2). The surge amplitude of 5 m is representative of motion in operation, while the 10 and 15 m amplitudes can be regarded as extreme cases, selected to enhance the motion effects.

All the cases with prescribed floater motion were simulated in below- and above-rated wind conditions, i.e., 8 m/s (tip-speed ratio $\lambda = 9.0$) and 15 m/s ($\lambda = 6.33$), respectively. The given λ values are for steady conditions, as the use of an active torque control makes the tip speed ratio change in time.

3.5 | Computational setup

In the present study, we have used the knowledge obtained from earlier discretization studies performed with MIRAS, for example, in Ramos-García et al.,^{25,27} in order to define the blade discretization. MIRAS-LL simulations employ a bound vortex discretized with 40 straight segments following a cosinus distribution, where the root and the tip region have a higher point density. The influence of the HAWC2 structural model fidelity of the blade on the response, loads, and stability of large turbines has been analyzed in Gözcü and Verelst.⁴⁶ A $28R \times 4R \times 4R$ Cartesian mesh has been employed in all cases, with a constant spacing of 3 m in the x, y, and z directions which adds up to a total of approximately 29 million cells. This cell size has been chosen to be equal to the one used to resolve the wake behind the complex multi-rotor turbine in van der Laan.⁴⁷ In that paper, the wake behind the Vestas multi-rotor concept was simulated and compared by different fidelity numerical models, including the MIRAS-FLEX5 coupling. The MIRAS-FLEX5 simulations used the same hybrid filament-particle-mesh solver as the present one obtaining a good agreement with the predictions of EllipSys3D LES-AL-FLEX5 for both mean velocities and turbulent intensity in the near and the far wake. A generally good agreement was also obtained with experimental short-range WindScanners data. The turbine hub is located at a position of 1.125R

TABLE 3 Test matrix for the prescribed harmonic pitch motion

	Floater pitch frequency [Hz]	Floater pitch period [s]	Floater pitch amplitude [deg]
Frequency study	0.01419	70.472	
	0.02838	35.236	6.38
	0.05676	17.618	
	0.11352	8.809	3.19
Amplitude study	0.02441	40.967	6.38
			12.7

TABLE 4 Test matrix for the prescribed harmonic surge motion

	Floater surge frequency [Hz]	Floater surge period [s]	Floater surge amplitude [m]
Frequency study	0.01419	70.472	
	0.02838	35.236	15
	0.05676	17.618	
	0.11352	8.809	5
Amplitude study	0.01419	70.472	10
			15

from the water level. So the distance from the ground to the blade tip when pointing straight down is 15m. All simulations have been carried out for 1,400 s of simulated time with a time step size of 0.025 s, which results in 56,000 time steps. Note that the chosen time step gives an azimuthal discretization under normal turbine operation of 0.8° to 1.1° , depending on the tip speed ratio. This means that approximately 450 time steps per revolution are simulated in below rated conditions while 330 time steps per revolution are simulated in the above-rated cases. In the extreme cases, where large instabilities are encountered, the time step is reduced by a factor of 2 to 0.0125 s. This temporal discretization is usually employed in HAWC2 stability analysis, giving accurate results even for the most dynamic load cases and instability studies as in Volk et al.⁴⁸ Each MIRAS simulation uses two-hundred 2.8 GHz processors on a Linux cluster, with an approximate computational time per case of 50 h. A free boundary condition is used in all directions but the ground plane, where a no-through condition is used.²⁸ An eight-order stencil is used to spatially discretize the vorticity equation. Moreover, a particle re-meshing every time step is used to maintain a smooth field, which is forced to remain divergence free by applying a periodic re-projection of the vorticity. The flow has been extracted every time step in 22 planes perpendicular to the flow direction (YZplanes), from 1R upstream to 20R downstream the rotor plane. Moreover, a single XZ plane is extracted. This plane contains the stream-wise direction and the height and is located at the rotor center. For reducing the necessary storage space, such planes are defined using 64×64 cells with a constant spacing and are written out as binary files. This means that in the YZ plane, the velocity extraction mesh is 2.5 times coarser than the computational mesh and in the XZ plane, the extraction mesh is 17.5 times coarser. Moreover, unless stated otherwise, only the last 700 s of the simulated time are used for the flow analysis, disregarding the first 700 s in order to avoid transients.

4 | SIMULATIONS OF THE IEA WIND 15 MW RWT MOUNTED ON THE WINDCRETE

An initial comparison between the HAWC2 standalone and the MIRAS-HAWC2 solvers is carried as a check of the aero-hydro-servo-elastic coupling. Three cases for the operating turbine are taken into account, in increasing degree of complexity: bottom-fixed turbine, floating turbine without waves, and floating turbine with waves. First, a comparative study between the solvers predictions of rotor integral and blade distributed quantities is carried out in what follows. Second, a comparison between the wake generated by a fixed and a floating turbine is presented. Note that all cases analyzed use the controller calibration described in Section 3.3.

4.1 | Turbine performance analysis.

Figure 3 depicts the mean in-plane and out-of-plane forces, F_x and F_y , respectively, and their standard deviation over the last 300 s of simulated time for winds speeds of 8 and 15 m/s, with tip-speed ratio of $\lambda = 9.0$ and 6.33, respectively. An excellent agreement is observed regarding the

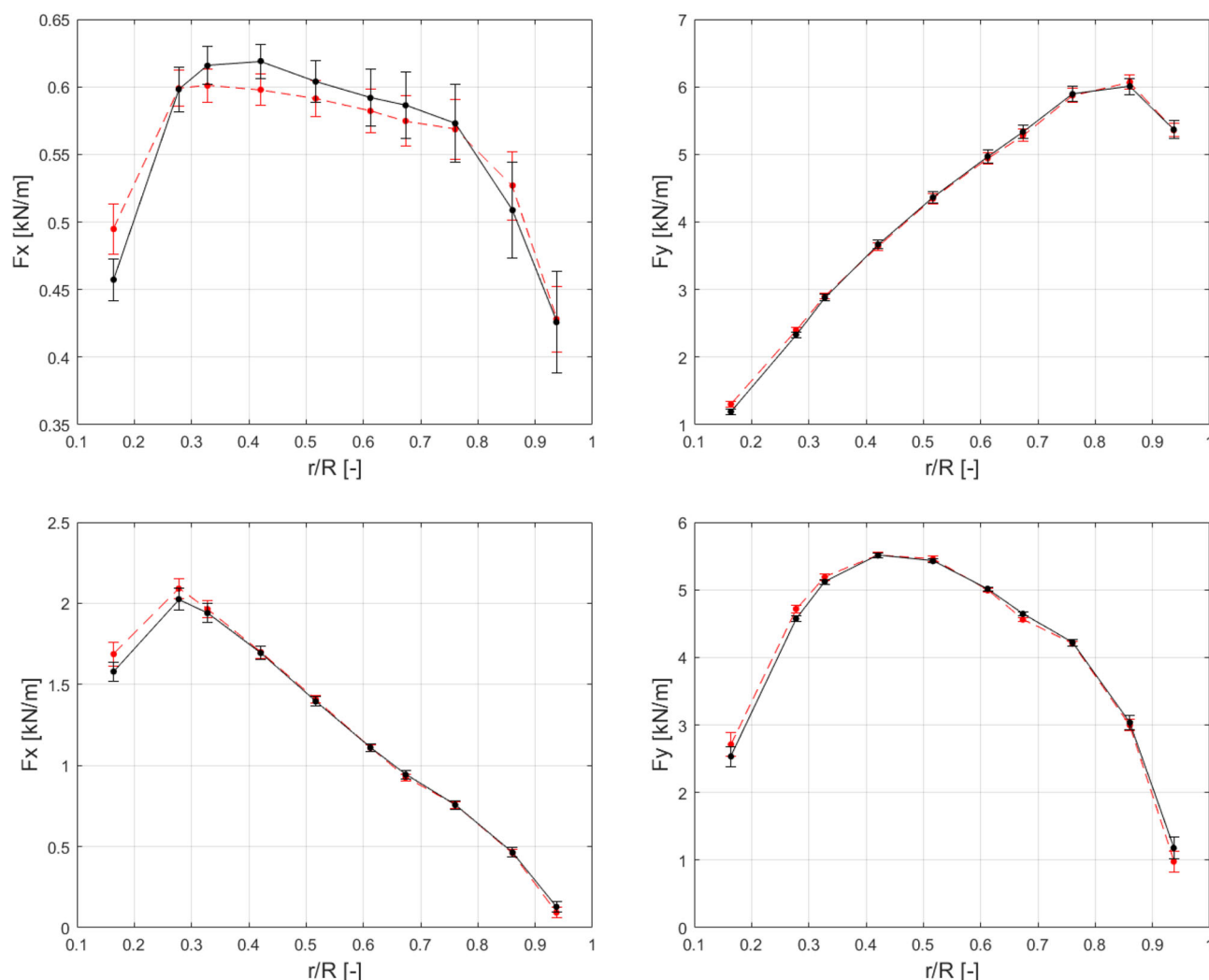


FIGURE 3 Mean and standard deviation of the in-plane and out-of-plane forces on blade one for the bottom-fixed IEA Wind 15 MW RWT. HAWC2-BEM in dashed red lines and MIRAS-HAWC2 in solid black lines (top, 8 m/s; bottom, 15 m/s)

out-of-plane force distribution in below-rated wind conditions, although larger relative differences appear in the in-plane component, especially around the root and mid-span regions. It is also remarkable that the LL method predicts a larger standard deviation of the loading in the tip vicinity. This is most probably related to the ground modeling in MIRAS, which is not accounted for in the lower fidelity solver. The proximity of the ground when the tip of the blades is pointing downwards creates a variation in the loading due to a ground effect. This has been confirmed by performing a simulation where the ground is not taken into account which shows a lower standard deviation of the loading. At 15 m/s, differences between the codes predictions of the in-plane force component are drastically reduced, while a good agreement is maintained for the out-of-plane one.

In terms of rotor performance predictions for the bottom-fixed IEA Wind 15 MW RWT, the BEM method under-predicts the mean power by a 1.4 %, while an excellent agreement is obtained for the thrust force at a wind speed of 8 m/s. At a wind speed of 15 m/s, differences in thrust and power between the codes are reduced to less than 0.1 %.

A comparison of the HAWC2-BEM and MIRAS-HAWC2 predicted forces and displacements in below-rated conditions without waves are shown in Figure 4. For clarity, only the last 700 s are depicted. A really good agreement is obtained between the codes in terms of the power and thrust variation in time. In general, the BEM method consistently under-predicts the mechanical power by less than 0.5% and slightly over-predicts the mean thrust force. Regarding the tower-top side-side (state pos x) and fore-aft (state pos y) displacements, there is a very small under-prediction of the first one and an over-prediction of the latter one by the BEM method probably related to the difference in thrust and aerodynamic damping.

At a wind speed of 15 m/s, Figure 5, differences in the time signals predicted by the solvers increase, although the computed mean values of all investigated quantities remain very similar. In this case, HAWC2-BEM consistently under-predicts both loads and displacements. Looking at

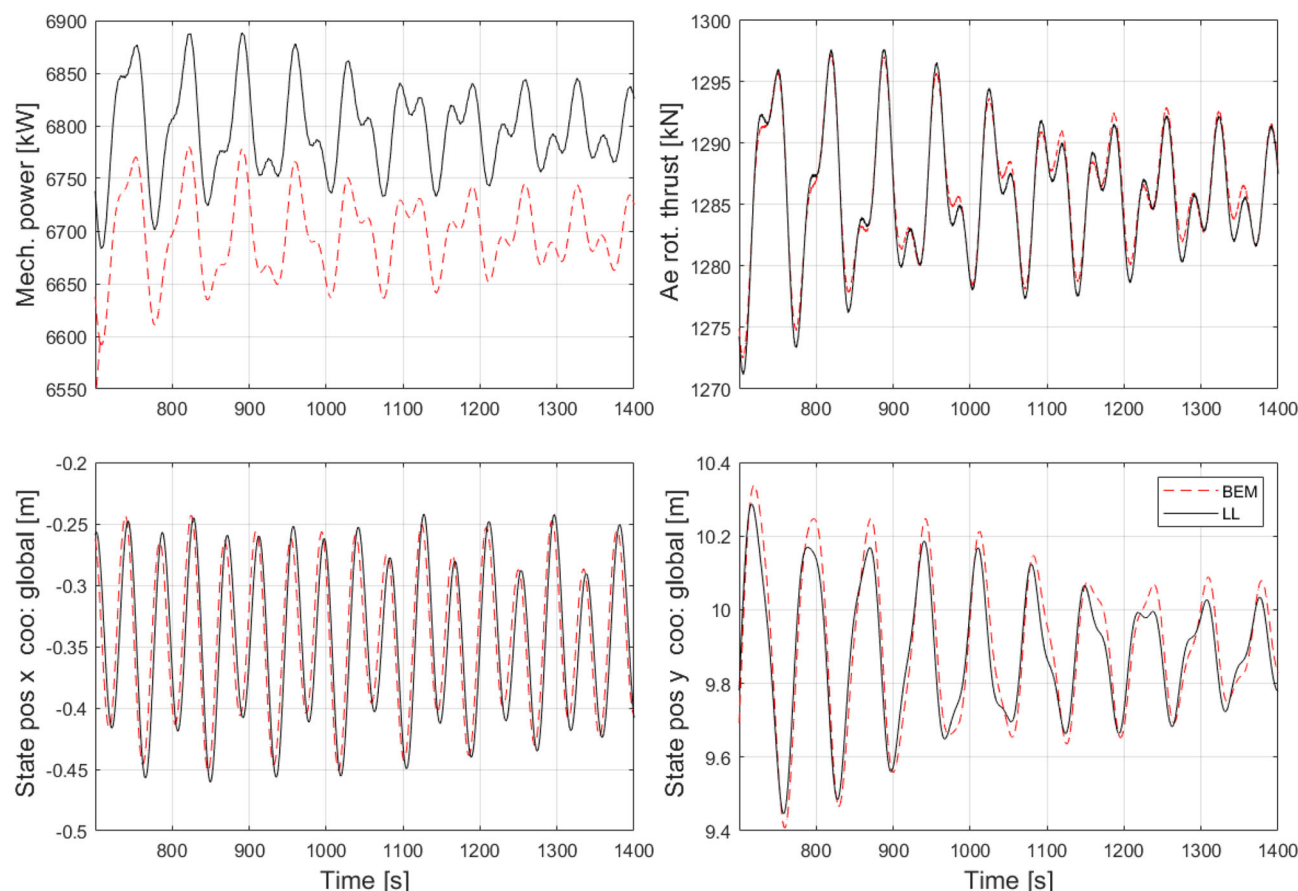


FIGURE 4 Floating IEA Wind 15 MW RWT mounted on the WindCrest platform without waves, comparison between BEM and LL simulations (8 m/s wind speed)

the spectral distribution of the thrust signal, Figure 6, there are some notable differences. The higher aerodynamic loads predicted by MIRAS leads to a dynamic behavior where the blade loading is more important in the overall turbine behavior, with the most important aerodynamic frequencies related to the blade passing, i.e., 3P and 6P, showing up in the spectral analysis of the thrust force. This frequencies are not clearly observed in the BEM simulations. In BEM, the power spectra density is dominated by the low frequency motion of the platform, over-predicting the peak value at the surge natural frequency of the floater (0.014 Hz) and under-predicting the pitch peak value (0.021 Hz).

A further comparison between the codes is carried out for multiple cases where regular waves are taken into consideration. The results obtained in below-rated conditions for the cases with 2 and 6 m Airy waves, with periods of 6 and 10 s, respectively, exhibited an excellent agreement between the codes (figures not included for the sake of brevity). However, a different picture is seen once the wind speed is increased to 15 m/s, as can be appreciated in Figure 7 where the turbine is subject to 2 m waves with a 6 s period. Differences between BEM and LL predictions increase considerably, with the BEM method predicting larger extreme loads, in terms of both the minimum and maximum values of thrust and power. This could be related to the different aerodynamic damping predicted by the codes since the system response at the floater natural frequencies is very sensitive to it. Moreover, the turbine undergoes larger oscillations in the BEM simulations, particularly in the fore-aft direction (state pos y), where the BEM method over-predicts the maximum amplitude of the motion by more than a 50% compared to the LL values. In general, it has been found that differences between BEM and LL simulations are more sensitive to variations in the motion frequency than in its amplitude, with BEM predictions deviating considerably from the LL predictions at high wave frequencies.

A spectral analysis of the thrust signal has been carried out to gain a better insight into the cases with regular waves at high wind speeds. As expected, the more distinct peak in the frequency domain appears exactly at the wave frequency, i.e., 0.16 and 0.10 Hz for the 2 and 6 m wave case, respectively. However, a striking difference between the codes is that the BEM simulations have a peak at the surge natural frequency, approximately at 0.014 Hz for the 2 m wave case, as can be seen in Figure 8 left. Since the wind is steady, and the wave is regular and at a frequency far from the surge natural one, there should be no load excitation, and the peak has to be due to a transient. It shows more in the BEM case since there is less aerodynamic damping and the transient survives for a longer time. An overall good agreement between the codes is obtained for the case with 6 m regular waves except for the low frequency range, as seen in Figure 8 right.

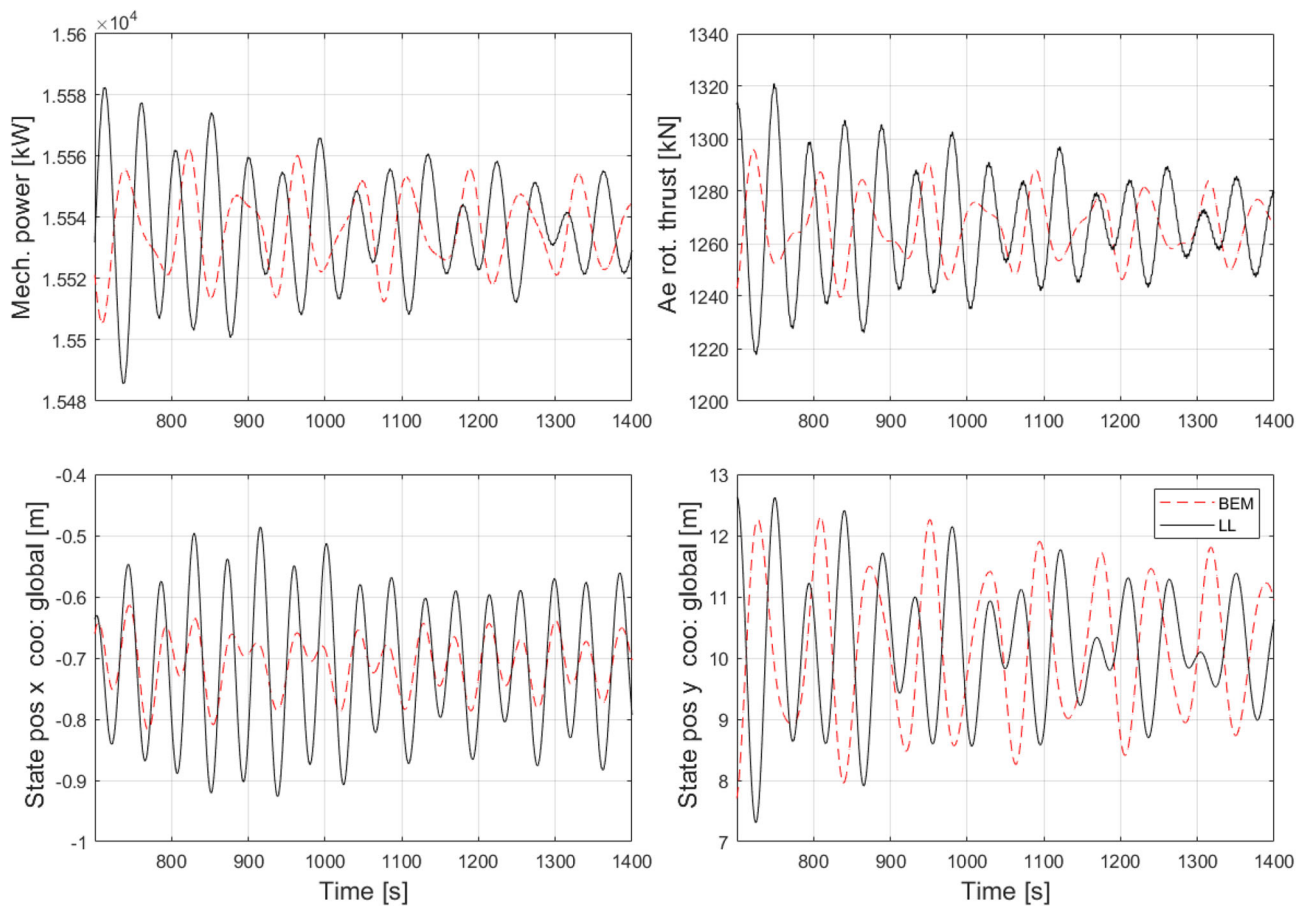


FIGURE 5 Floating IEA Wind 15 MW RWT mounted on the WindCrest platform without waves, comparison between BEM and LL simulations (15 m/s wind speed)

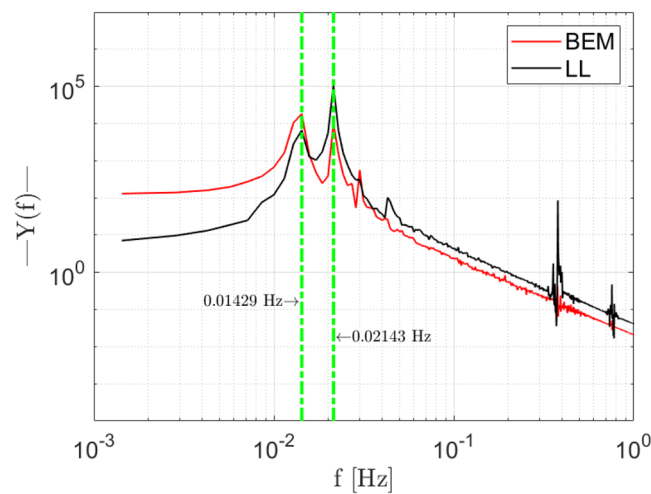


FIGURE 6 Spectral analysis of the thrust signal predicted for the floating IEA Wind 15 MW RWT mounted on the WindCrest platform without waves. Comparison between BEM and LL simulations (15 m/s case)

4.2 | Wake analysis

A comparison of the wakes generated by the bottom-fixed and the floating IEA Wind 15 MW RWT in below-rated conditions is presented in what follows. A low wind speed case is selected because it features the largest differences between the wakes. At 8 m/s, the turbine is operating at an

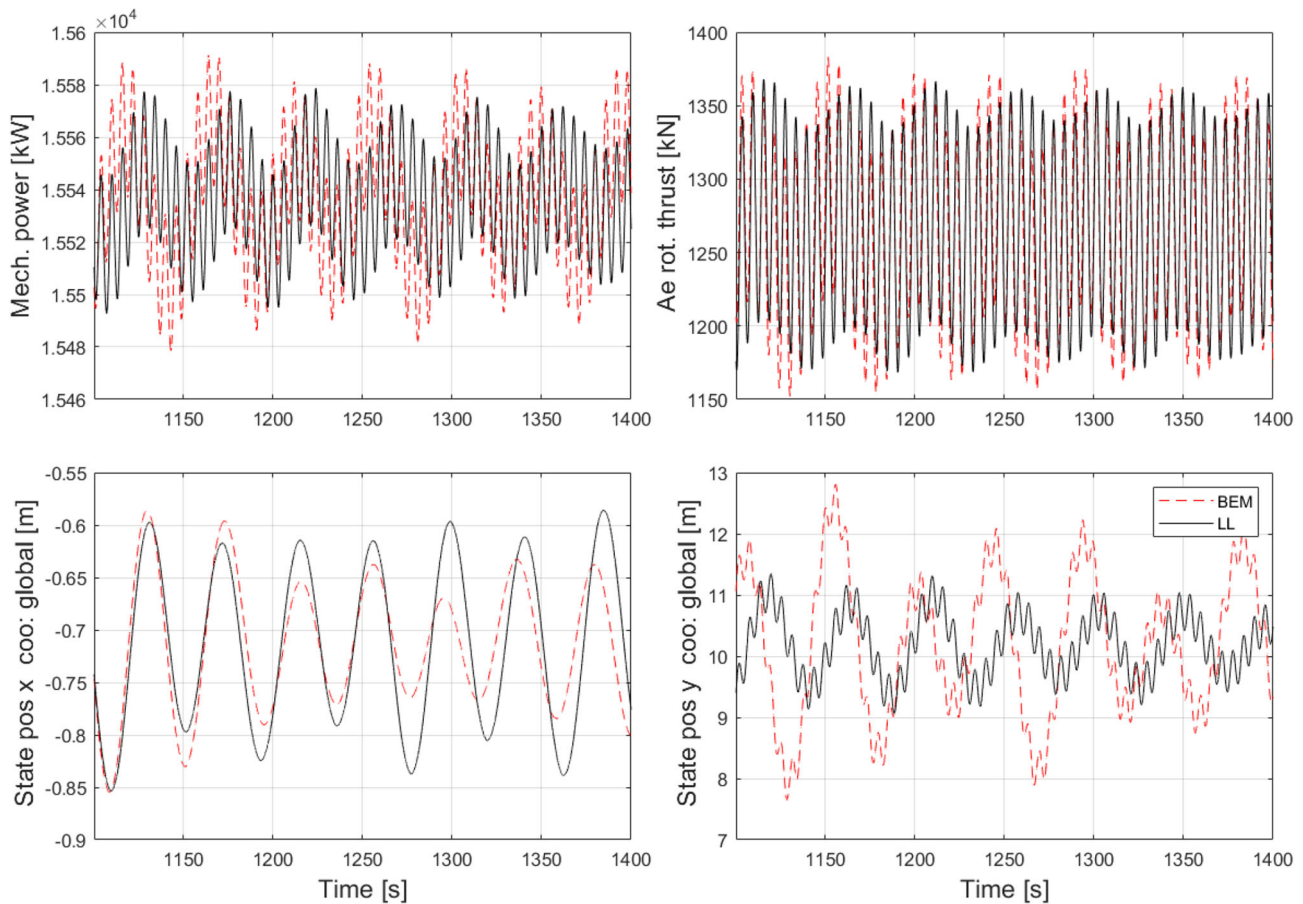


FIGURE 7 Floating IEA Wind 15 MW RWT mounted on the WindCrest platform and subject to regular waves with 2 m height and period of 6 s. Comparison between BEM and LL simulations (15 m/s case)

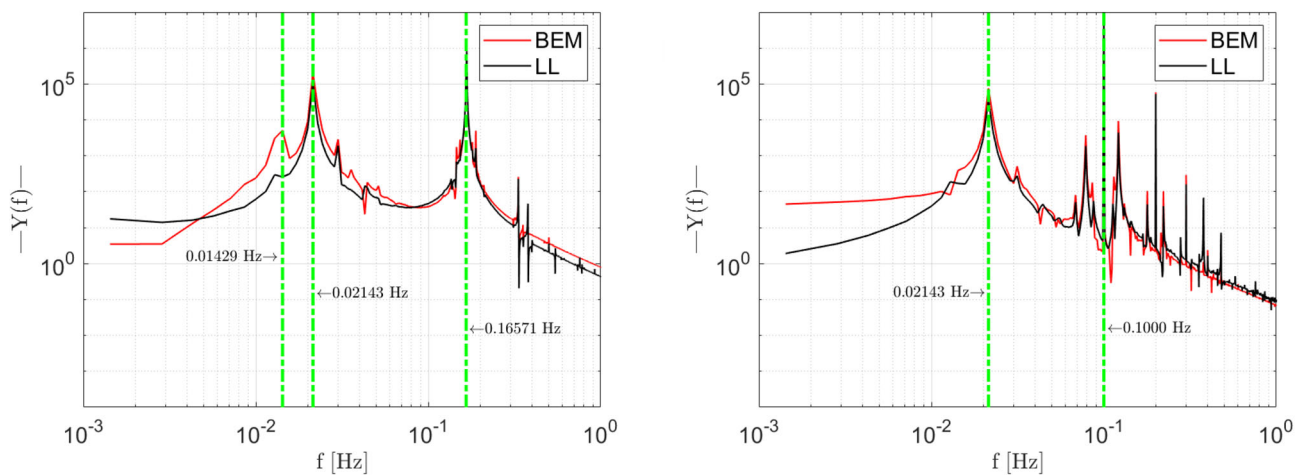


FIGURE 8 Spectral analysis of the thrust signal predicted for the floating IEA Wind 15 MW RWT mounted on the WindCrest platform and subject to regular waves with (left) 2 m height and period of 6 s and (right) 6 m height and period of 10 s. Comparison between BEM and LL simulations (15 m/s case)

optimal tip speed ratio for extracting maximum power. Note that only small differences have been observed when comparing the wake of the floating cases with and without waves (for the considered cases) and, therefore, such comparison is not presented here.

Figure 9 shows stream-wise velocity and turbulence intensity contours at multiple downstream locations, i.e., 2, 4, 6, 10, 14, 16, 18, and 20R downstream the rotor plane. The wake generated by a bottom-fixed turbine with a non-turbulent inflow, depicted in the top sub-figures, remains

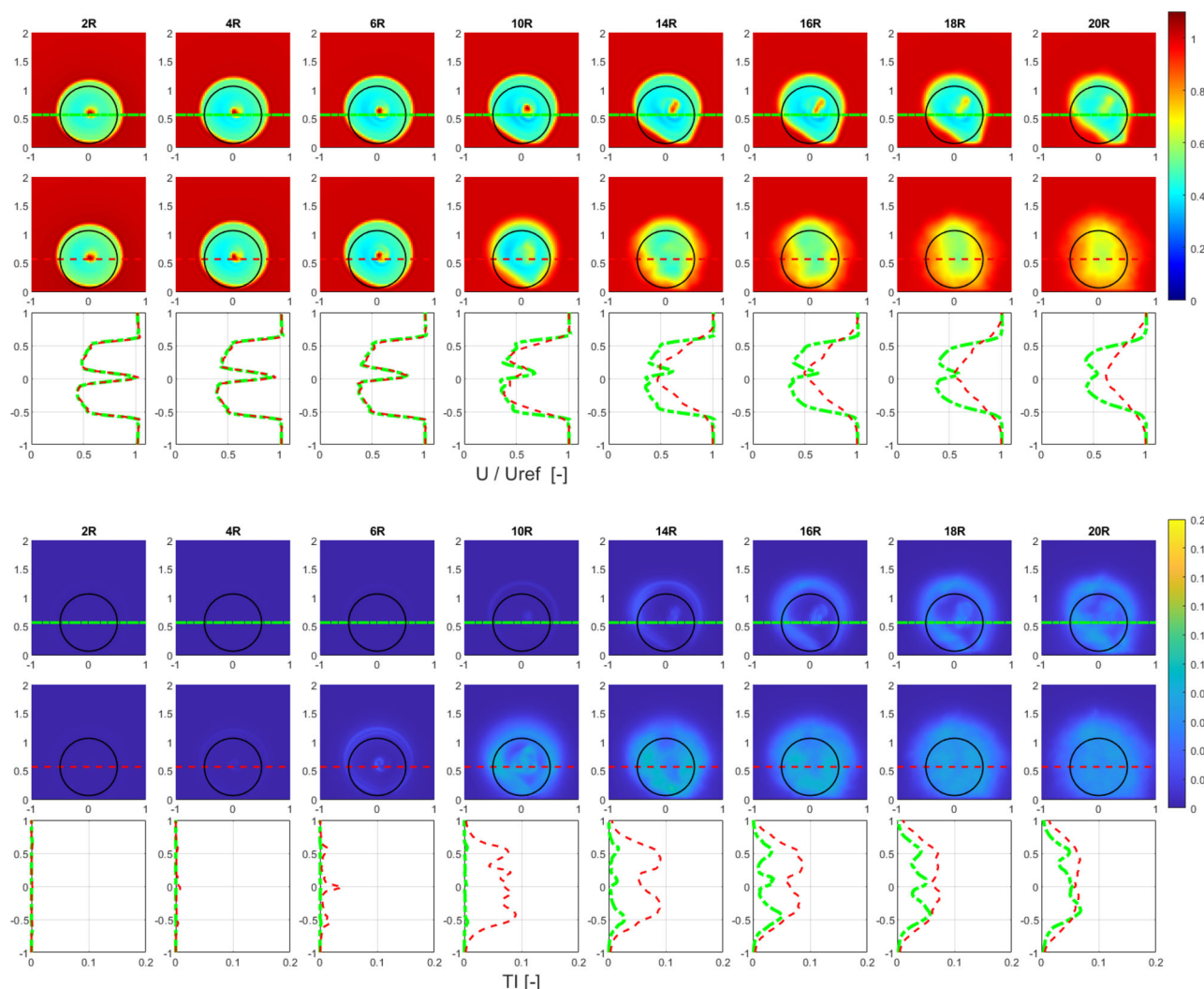


FIGURE 9 Comparison between the (top-figure) mean stream-wise velocity and (bottom-figure) turbulent intensity in the wakes generated by the bottom-fixed and the floating IEA Wind 15 MW RWT at different downstream distances from the rotor plane. Mean stream-wise velocity and turbulence intensity contours on multiple YZ planes downstream the rotor plane (top sub-figures) bottom-fixed case and (mid subfigures) floating case. (bottom sub-figures) profiles along the horizontal marked line drawn at hub height where green represents the bottom fixed case and red depicts the floating one

undisturbed for a longer distance compared to the one generated by a floating turbine, which is depicted in the mid sub-figures. The later one reaches a Gaussian distribution between 10 to 14 R downstream the rotor plane. However, it is observed that for the wake generated by the bottom-fixed turbine the Gaussian shape is first reached further downstream than 20 R . This can be further explained by looking at the turbulence intensity contours. The floating turbine generates more mixing in the flow, developing a higher turbulence level which helps promoting a faster break down of the strong vortex structures. Note that at far downstream positions the turbulence intensity profile of both wakes becomes very similar. Contrary to what is observed in the present work Kopperstad et al,¹⁹ spar-buoy concept showed a quite similar wake recovery as in the bottom fixed case.

In an effort to define a set of simplified quantities that can describe the wake characteristics, we introduce the wake velocity recovery, Equation 6, and the integrated turbulence intensity, Equation 7. These terms are used throughout the manuscript to characterize the wake in a simple manner. Figure 10 depicts such quantities as function of the downstream distance from the rotor plane for the bottom fixed and the floating cases. Note that the area of integration is defined by the rotor disk excluding the hub region, which is approximately 0.1 R . This figure confirms what was already shown in the previous analysis, the faster recovery of the wake of a floating wind turbine in laminar inflow conditions, which is tied to the higher TI level produced by the motion of the floater. For example, at a distance of 12 R , the wake recovery is increased from 0.47 for the bottom fixed case to 0.56 for the floating case.

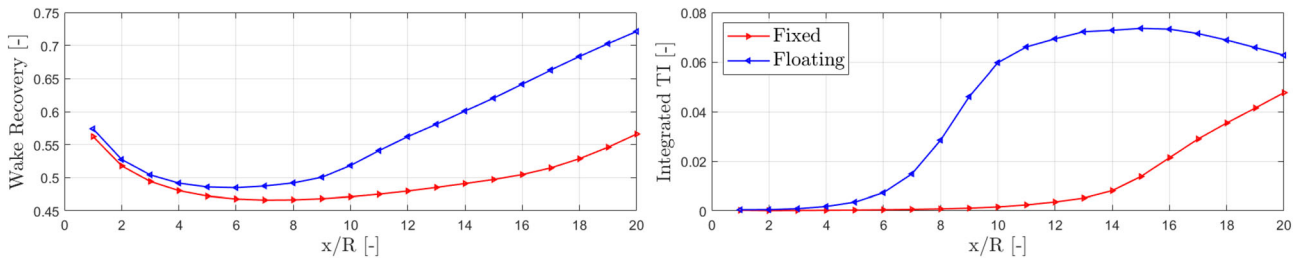


FIGURE 10 Integrated wake quantities for the bottom-fixed and floating IEA Wind 15 MW RWT at a wind speed of 8 m/s. (left) Wake velocity recovery; (right) integrated turbulent intensity

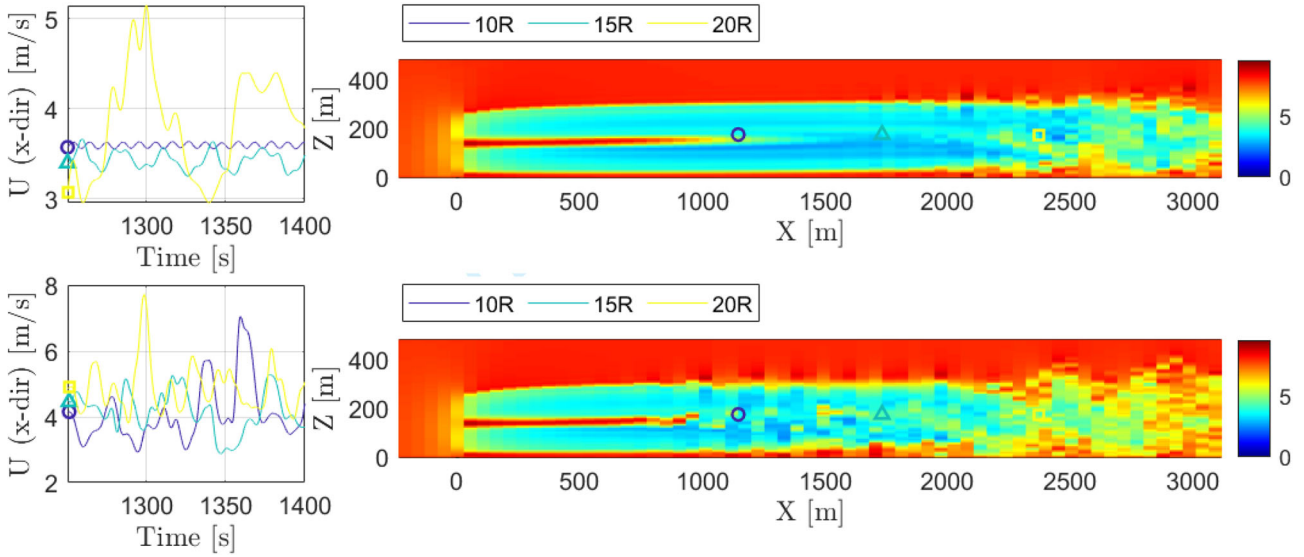


FIGURE 11 (Left) Time variation of the stream-wise velocity at three different downstream locations; (right) instantaneous stream-wise velocity contours on a XZ plane crossing the turbine center at the simulation time of 1,250 s. Comparison between (top) the bottom-fixed and (bottom) the floating IEA Wind 15 MW RWT

$$Wake Recovery = \frac{\int_A U dA}{\int_A U_0 dA}, \quad (6)$$

$$Integrated TI = \frac{\int_A TI dA}{A}. \quad (7)$$

Figure 11 shows the time variation of the hub height stream-wise velocity for the last 150 s of simulated time at three different downstream locations 10, 15, and 20 R and the instantaneous stream-wise velocity contours in an XZ plane crossing through the center of the rotor at the simulation time of 1,250 s. This figures gives a very good overview of the floater influence in the wake development. Large differences can be seen between the velocity contours of both cases and more especially between the time signals at the 10 and 15 R locations, with pronounced low-frequency oscillations for the floating case.

5 | INVESTIGATION OF THE EFFECT OF MEAN TILT IN THE WAKE DEVELOPMENT

Due to the mean aerodynamic thrust on the rotor, a mean operational pitch angle should usually be considered for floating wind turbines. This results in a larger operating mean tilt angle, when compared to their bottom-fixed counterparts. In order to understand how the mean rotor tilt angle affects the wake development, stream-wise velocity contours for different downstream located YZ planes behind the bottom-fixed IEA Wind 15 MW RWT with 0°, 10°, 15°, and 30° of tilt are presented in Figure 12. From the figure, it is clear that a positive tilt angle, defined with

the rotor facing upwards, deflects the wake towards higher altitudes. This is consistent with the findings of Wise and Bachynski,⁴⁹ who also pointed out the similarity with the well-known horizontal deflection from yaw-misalignment.

It is also very interesting to see how the wake curls up forming a mushroom shape in its cross-sectional cut for large tilt angles, which further helps bending the wake away.

To more easily quantify the wake properties for the different tilt cases, Figure 13 shows the computed wake recovery and the integrated turbulence intensity, computed following Equations 6 and 7. From the figure, it is clear that the larger the tilt angle, the faster the wake recovers. Note that with a tilt angle of 30° , almost 90% of the wake is recovered at the 20R downstream location. We also note that this tilt angle is not realistic for a floating wind turbine, but it is included here for completeness. On top of the upwards motion of the wake, the wake stretching towards a mushroom shape breaks down the cylindrical wake form and promotes its recovery. Differences between the different tilt cases are already notable at the closest downstream station in the plot (1 R), and they increase with the downstream distance. For example at $x/R = 12$, a change from 0° to 10° tilt increases the wake recovery from 0.45 to 0.52. In the far wake, the differences start to decrease. Regarding the integrated TI, it is interesting that the 30° tilt case shows the higher values up to 14 R, after that there is an inversion of the trend and this case has the lowest integrated TI in the far wake. This effect is mostly related to the wake moving away from the integrated area and not to the wake dissipation itself.

6 | INVESTIGATION OF THE EFFECT OF FLOATER HARMONIC MOTION IN PITCH

In order to isolate the aerodynamic effects related to the different DoFs, we first force a prescribed harmonic pitch motion of the floater, which is a very relevant DoF for a floating wind turbine from an aerodynamic point of view. Note that all cases presented use the DTU Wind Energy

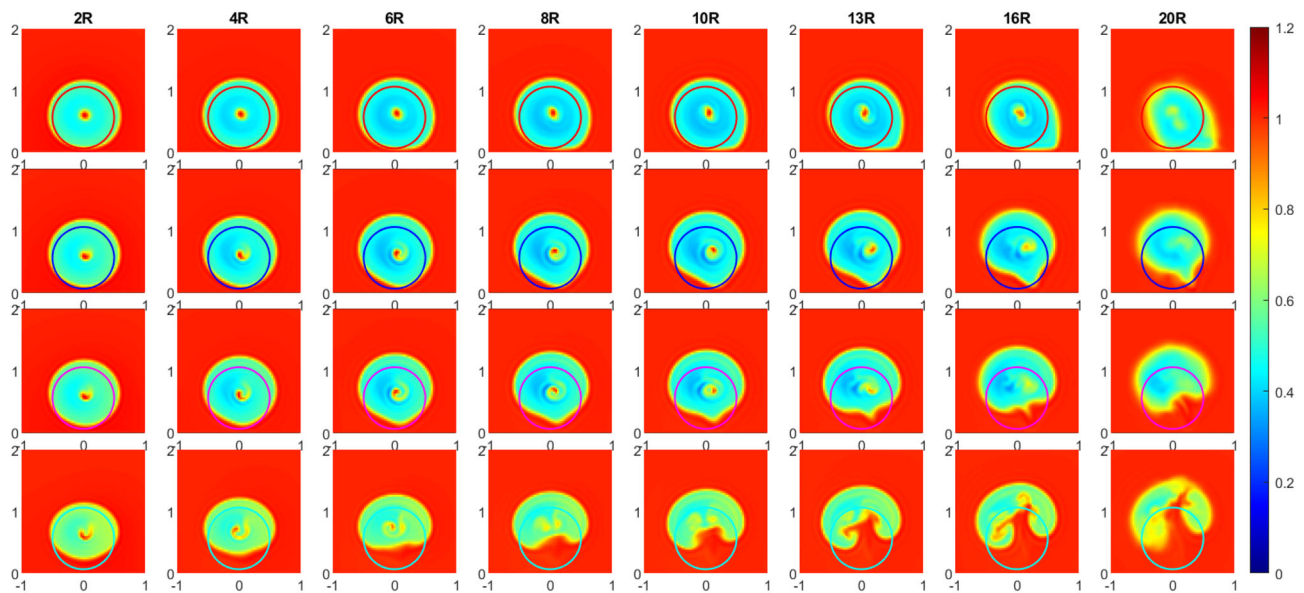


FIGURE 12 Mean stream-wise velocity contours on different YZ planes downstream the rotor plane for the bottom-fixed IEA Wind 15 MW RWT with tilt angles (from top to bottom) 0° , 10° , 15° , and 30° at a wind speed of 8 m/s

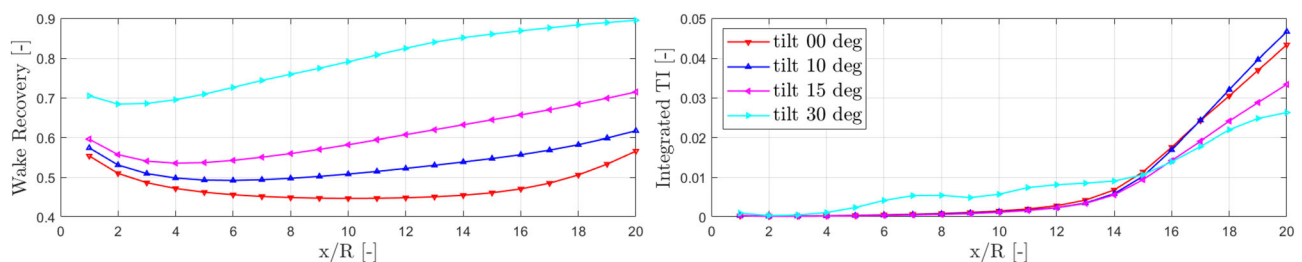


FIGURE 13 Integrated wake quantities for the bottom-fixed IEA Wind 15 MW RWT with 0° , 10° , 15° , and 30° tilt angles at a wind speed of 8 m/s. (Left) Integrated wake velocity recovery; (right) Integrated Turbulent Intensity

Controller calibrated as described in Section 3.3. All cases are simulated in below- and above-rated conditions, i.e., wind speeds of 8 and 15 m/s, respectively.

The test matrix for the harmonic pitch motion is presented in Table 3. The investigation is divided in five parts. First, a turbine load performance analysis is carried out. Key turbine quantities like rotor rotational speed, blade pitch, mechanical power, and aerodynamic thrust are presented and analyzed for the last pitching cycle of the floater. We have chosen to present the mechanical power at the shaft and not the aerodynamic one since the latter is partially used to accelerate the inertia of the rotor and such contribution is not felt at the shaft. The depicted power is therefore calculated as the shaft moment multiplied by the rotational speed. This study is combined with a qualitative wake analysis where three-dimensional iso-vorticity contours are investigated for six different time instants during the last pitching cycle to cover the complete turbine oscillation, i.e., t_1 , t_2 , t_3 , t_4 , t_5 , and t_6 . Ring distributions of the in-plane and out-of-plane blade forces are shown for selected cases in order to better understand the turbine behavior under such dynamic conditions. Second, a similar study is carried out for the extreme frequency cases. Such cases display a very different turbine behavior compared to the lower frequency ones, and it is therefore for the sake of clarity that they are presented in a different section. Third, an analysis of the different rotor states along the last pitching cycle is carried out in order to further characterize the aerodynamic behavior of a floating turbine. Fourth, a comparative study of the maximum blade root and tower base bending moments for all simulated cases is shown. Fifth and last, a quantitative wake study is performed with focus on the wake recovery.

6.1 | Turbine behavior and iso-vorticity comparison in pitch

The variation of key turbine quantities over the last pitching cycle for three different frequencies in below-rated conditions is presented in Figure 14. MIRAS-HAWC2 results are compared to the standalone HAWC2 simulations with the in-built BEM aerodynamic module. The BEM solver tends to under-predict the maximum attained power and thrust for the low and mid frequency cases. Such maximum values appear in the second half of the forward motion of the floater as it enters negative floater pitch values. For example, at the frequency of 0.01419 Hz, the power is under-predicted by approximately a 10%. However, a different trend is observed for the 0.05676 Hz case. For this case, the pitch control is active in the last half of the forward motion and beginning of the backwards one which could explain the change in tendency. Regarding the minimum loading region, there is a general tendency for the BEM method to slightly under-predict the thrust during the first part of the backwards motion, $\approx t_6 - t_1$, and considerably over-predict it during the second part, $\approx t_1 - t_2$. It is in this region where the relative error between the code

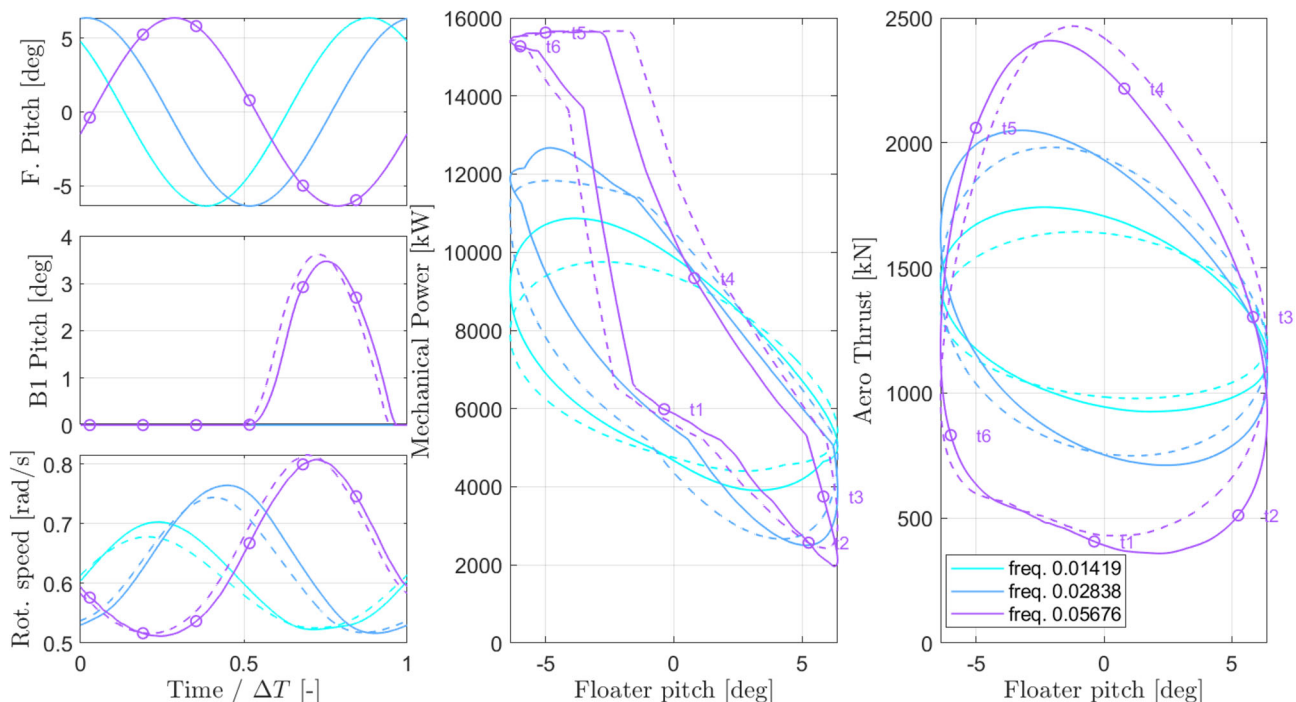


FIGURE 14 Aerodynamic quantities for the last cycle of a prescribed pitching motion of the floater with an amplitude of 6.38° and different frequencies. Wind speed of 8 m/s. Solid lines represent MIRAS-HAWC2 simulations, while dashed lines represent HAWC2-BEM

predictions is largest. As it shown in what follows, this is a sign of the BEM method not accurately capturing the rotor aerodynamics in local vortex ring state, observed between the times t_1 and t_2 .

The vorticity iso-contours of the 0.05676 Hz case for the six different marked times are depicted in Figure 15. This case is chosen because it is the highest frequency case in the set and it shows the most distinguishable wake effects. It can be appreciated in the figure how at t_1 and t_2 the upward pointing blades are striking the wake generated by the previous passing blade, at the same time as the two tip vortices merge into one, increasing its total thickness. This is an initial evidence of local vortex ring state isolated to the tip region. Note that the large flexibility of the IEA Wind 15 MW RWT blades could have a strong impact in its aerodynamic behavior, with the outer part of the blades bent forward and making them less prone to hit their own wake during the backwards motion, and vice-versa during the forward motion. Focusing on the iso-vorticity contour at t_3 , it is clearly appreciated how once the turbine starts moving forward the thicker tip vortex (formed by a combination of two) splits back into two separate ones. For clarity, a red box is drawn to highlight these effects for times t_1 , t_2 , and t_3 . Note that for the rest of the selected times, the near wake does look more conventional although one can discern how the merged tip vortices travel downstream.

Figure 16 shows a ring distribution of the MIRAS predicted F_x and F_y forces in blade one. Sub-figures to the left represent the force distributions over the first half of the last pitching cycle; meanwhile, sub-figures to the right depict the second half. The colored outer ring represents the direction of the floater motion, red represents backward motion, and blue features the forward one. Moreover, the sub-figure on the top of each ring distribution shows the variation of the floater pitch and the root bending moments for blade one over the non-dimensionalized time, $(t - t_0)/(t_f - t_0)$. Here, t_0 is the start time of the last pitch cycle, and t_f is the final time. M_x is the flap-wise root bending moment, and M_y is the chord-wise root bending moment. Finally, the small gray band in the top sub-figures during the second half of the pitch cycle highlights simulated time non-included in the ring distribution.

During the backwards motion of the floater, approximate time from 0.00 to 0.17, the in-plane forces reach a plateau near zero loading along the span. The out-of-plane force displays slightly larger values near the tip and small negative values in the mid and in-board regions, which is yet another indication of the vortex ring state. This can be explained by the gathering of vortex structures in the root region which induces a negative inflow velocity at the blade. Focusing on the root bending moments, it is interesting that the chord-wise component, M_y , reaches a maximum during the transition from backward to forward motion of the turbine and vice-versa, non-dimensional times of 0.29 and 0.79, respectively. Regarding the flap-wise component, M_x , its maximum absolute value appears in accordance with the maximum out-of-plane blade loading, attained between non-dimensional times of 0.63 and 0.67. The minimum value of M_x is predicted during the backward motion, non-dimensional time of 0.08, and it is again in accordance with the minimum out-of-plane blade loading.

In what follows, we present the frequency study in above-rated wind conditions. Key quantities are shown in Figure 17. As expected, the blade pitch control is active in all cases. In general, it is seen that differences between the codes predictions are large for the 0.05676 Hz case while they are quite small in the lower frequency cases. Focusing on the 0.05676 Hz case, the LL simulations predict a speed up of the rotor due to the generator servo action while maintaining a lower blade pitch, which adds a time offset in comparison to BEM. In this case, the predicted power is quite different between the codes. Especially during the backwards motion of the floater, $\approx t_6 - t_1 - t_2$, where the BEM method tends

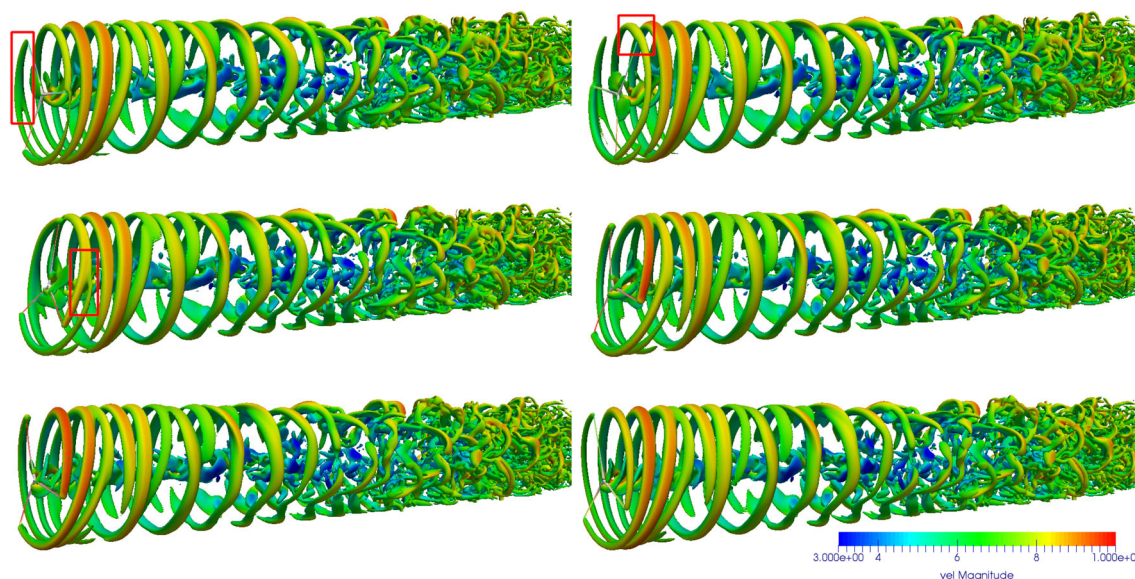


FIGURE 15 Three-dimensional vorticity contours at six selected times over the last cycle of a floater prescribed pitch motion with 6.38° amplitude and a frequency of 0.05676 Hz. From left to right and top to bottom, t_1 , t_2 , t_3 , t_4 , t_5 , and t_6 . Wind speed of 8 m/s

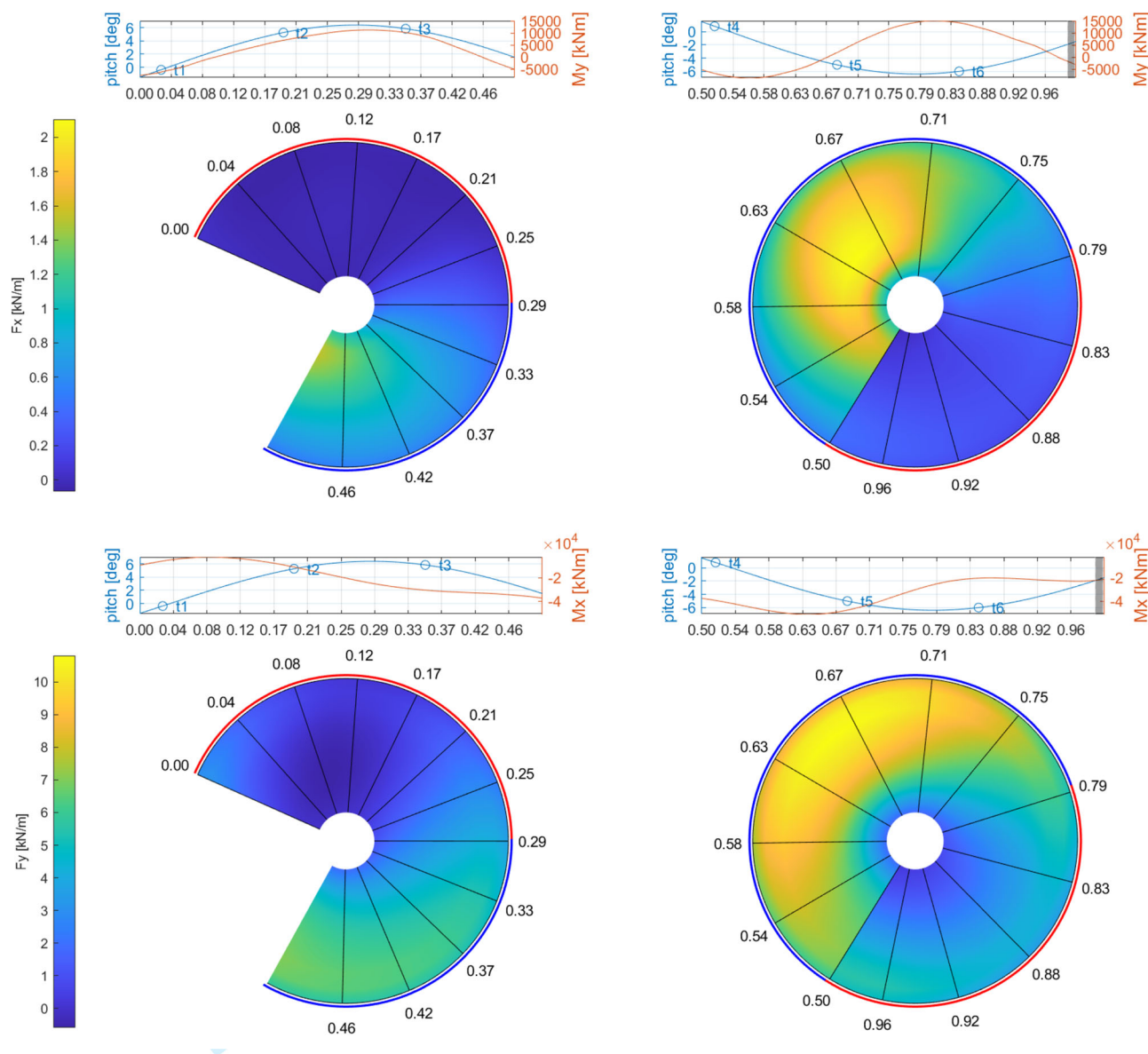


FIGURE 16 Annular distribution of the (top) in-plane force, F_x , (bottom) out-of-plane force, F_y , for the last cycle of a prescribed pitch motion of the floater with an amplitude of 6.38° and a frequency of 0.05676 Hz. Wind speed of 8 m/s. Blade 1

to consistently under-predict the power up to a 9 % respect to LL at t_2 . Differences in thrust are large, with a clear under-prediction of the maximum thrust by the BEM code when the floater starts moving forward, $\approx t_3$, and a constant offset between the codes during the complete backwards motion, $\approx t_6 - t_1 - t_2$.

Analyzing the iso-vorticity contours for the 0.05676 Hz case in above-rated wind conditions, depicted in Figures 18, there are clear differences compared to the 8 m/s case. The blade tip is not that highly loaded, and therefore, the tip vortex is not as concentrated; it seems to be distributed along a larger span region. This is highlighted with a red box in the t_3 snapshot. We thus emphasize that in this case, the blades are far from entering local vortex ring state. As it is shown in Section 6.3, the rotor remains in normal turbine operation state during the complete cycle. It is remarkable that during the forward motion of the floater the maximum out-of-plane blade loading appears in the mid-span region, as shown in the ring distributions of Figure 19. This creates a particular span-wise variation of the bound vortex circulation which produces a double tip vortex structure, which can clearly be appreciated in the t_4 , t_5 , and t_6 iso-vorticity snapshots. Note that during the forward motion of the floater all blade tips are subjected to negative F_x and F_y loads. This is due to the servo action, pitching the blades towards feather, combined with the effect of blade torsion, which helps deloading the tip by reducing the effective angle of attack.

We next investigate the amplitude dependence of the floater pitch motion at the fixed frequency of 0.024421 Hz. Pitching amplitudes of 3.19° , 6.38° , and 12.76° are considered in below-rated wind conditions. Key selected sensors are shown in Figure 20 for the BEM and LL

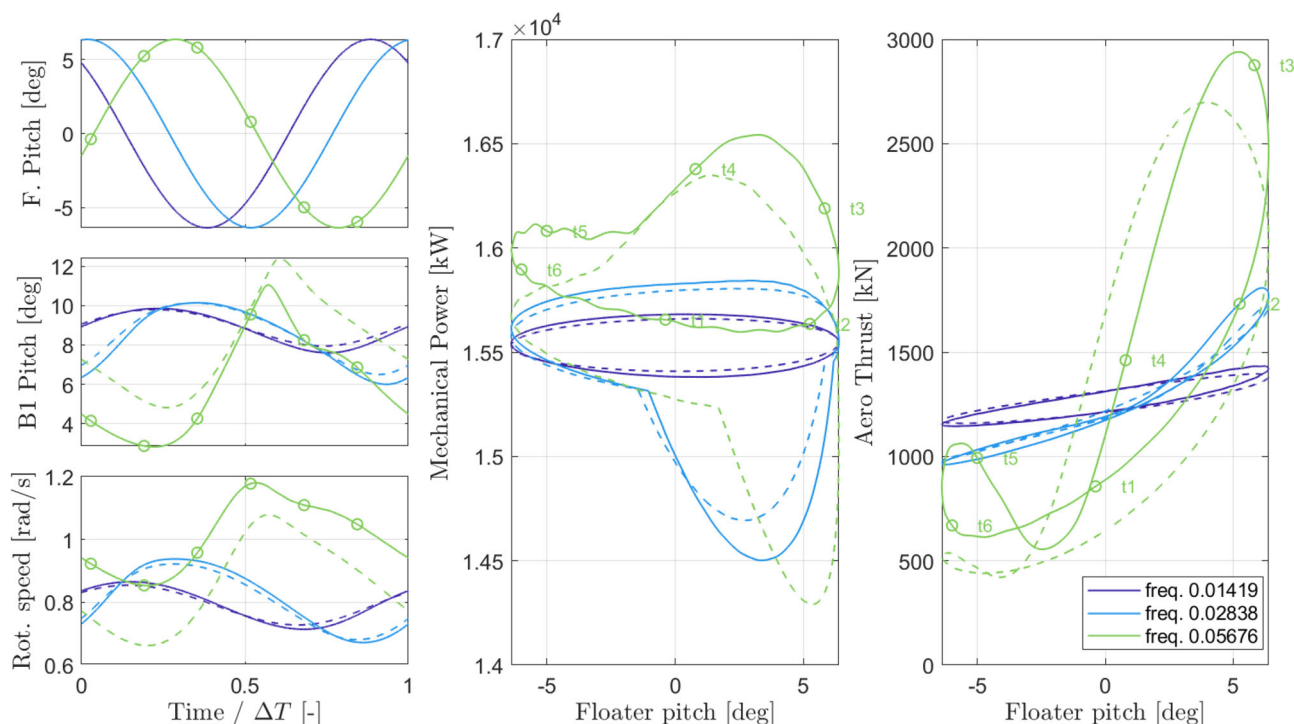


FIGURE 17 Aerodynamic quantities for the last cycle of a prescribed pitching motion of the floater with an amplitude of 6.38° and different frequencies. Wind speed of 15 m/s. Solid lines represent MIRAS-HAWC2 simulations, while dashed lines represent HAWC2-BEM

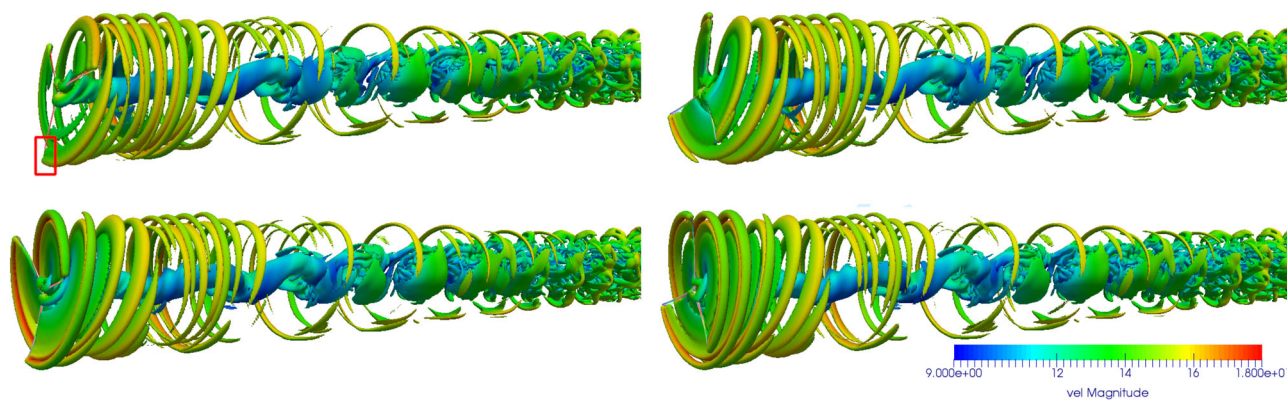


FIGURE 18 Three-dimensional vorticity contours at six selected times over the last cycle of a floater prescribed pitch motion with 6.38° amplitude and a frequency of 0.05676 Hz. From left to right and top to bottom, t_3 , t_4 , t_5 , and t_6 . Wind speed of 15 m/s

simulations. It is interesting here to investigate the change in shape of the load loop for the 12.76° case compared to the lower amplitudes motions. This change is similarly captured by both solvers although small differences in thrust between t_6 and t_1 can be linked to a more aggressive blade pitch actuation in the LL simulations. Another remarkable difference between the solver predictions appears during the second half of the backward motion of the floater, $\approx t_3 - t_4$, which, judging from the iso-vorticity contours presented in Figure 21, is related to the upwards pointing blade entering its own wake in what can be identified as local vortex ring state. This phenomenon is highlighted with a red box in the t_3 and t_4 iso-vorticity snapshots.

Results for the different amplitude cases at the wind speed of 15 m/s are presented in Figure 22. The largest differences between the codes are observed in the 12.76° amplitude case. The maximum thrust is strongly under-predicted by the BEM code by a 10 %, while the minimum power is over-predicted by a 3 %. It also is interesting that during the beginning of the forward motion of the floater, $t_5 - t_6$, there appears a bump in the power and thrust. This is mostly related to the action of the controller switching quickly from below-rated to above-rated regulation,

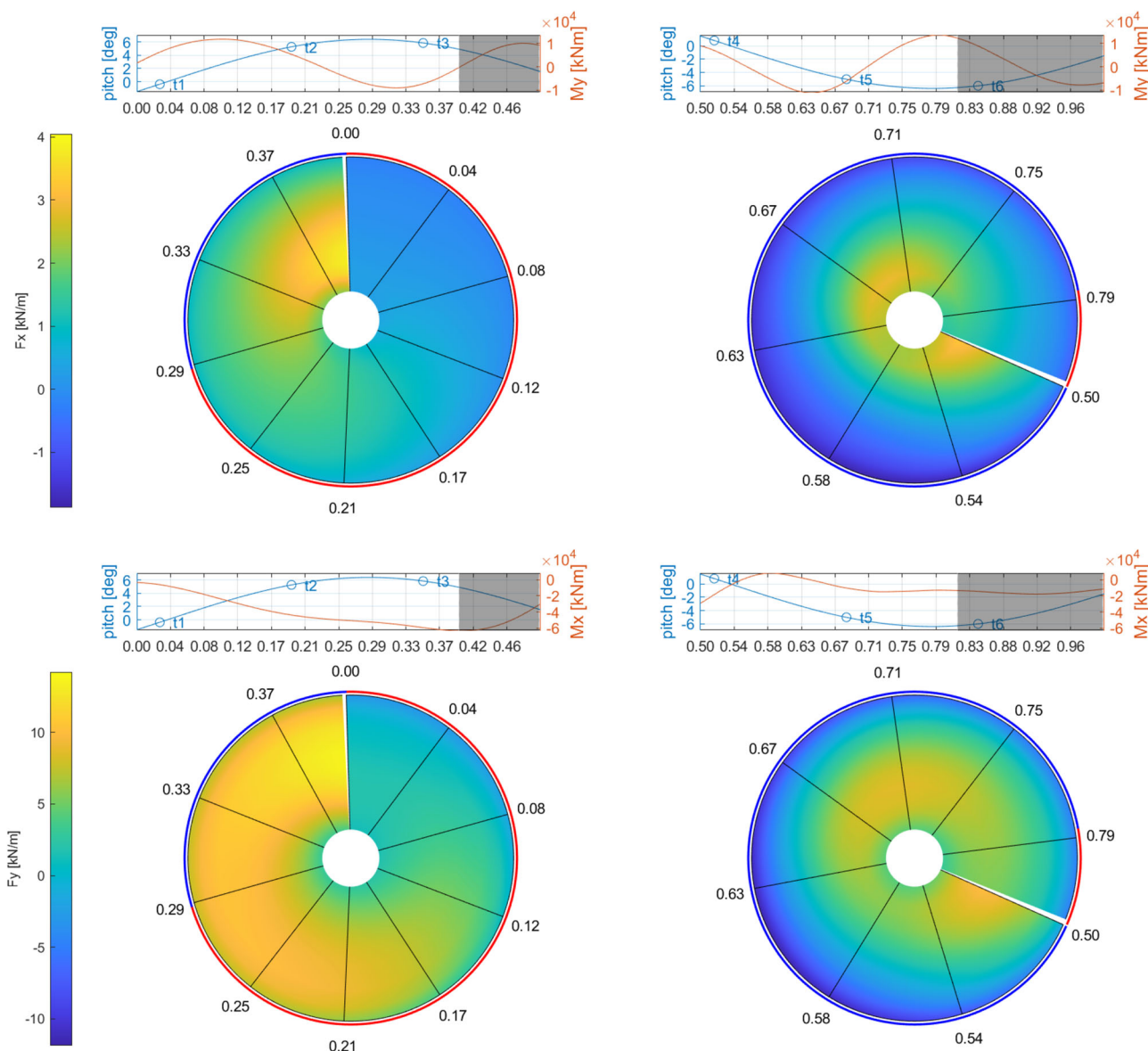


FIGURE 19 Annular distribution of the (top) in-plane force, F_x , and (bottom) out-of-plane force, F_y , for the last cycle of a prescribed pitch motion of the floater with an amplitude of 6.38° and a frequency of 0.05676 Hz. Wind speed of 15 m/s. Blade 1

which increases the rotational speed and pitches the blade towards stall. This sudden change is predicted by both codes. However, the LL predicts a steeper response, probably due to the fact that it predicts larger load variations.

For the sake of brevity, and since they do not provide new insight in our investigation, the iso-vorticity contours for these cases are not included.

6.2 | Turbine behavior and iso-vorticity comparison for extreme cases in pitch

As it is shown in the analysis of maximum bending moments, Section 6.4, the cases where the floater motion is prescribed with a frequency of 0.11352 Hz, representative of the wave loads in harsh sea states, and an amplitude of 6.37° displays the most extreme loads. These cases are presented in a different section in order not to mask the analysis of the moderate cases. It is important to mention that, in order to take a less conservative approach in the simulation of such extreme cases, the controller over-speed thresholds that trigger the safety system and the wind turbine shut-down were deactivated. Such thresholds would trigger an automatic shut down of the turbine after a few seconds in order to prevent damages to the machine.

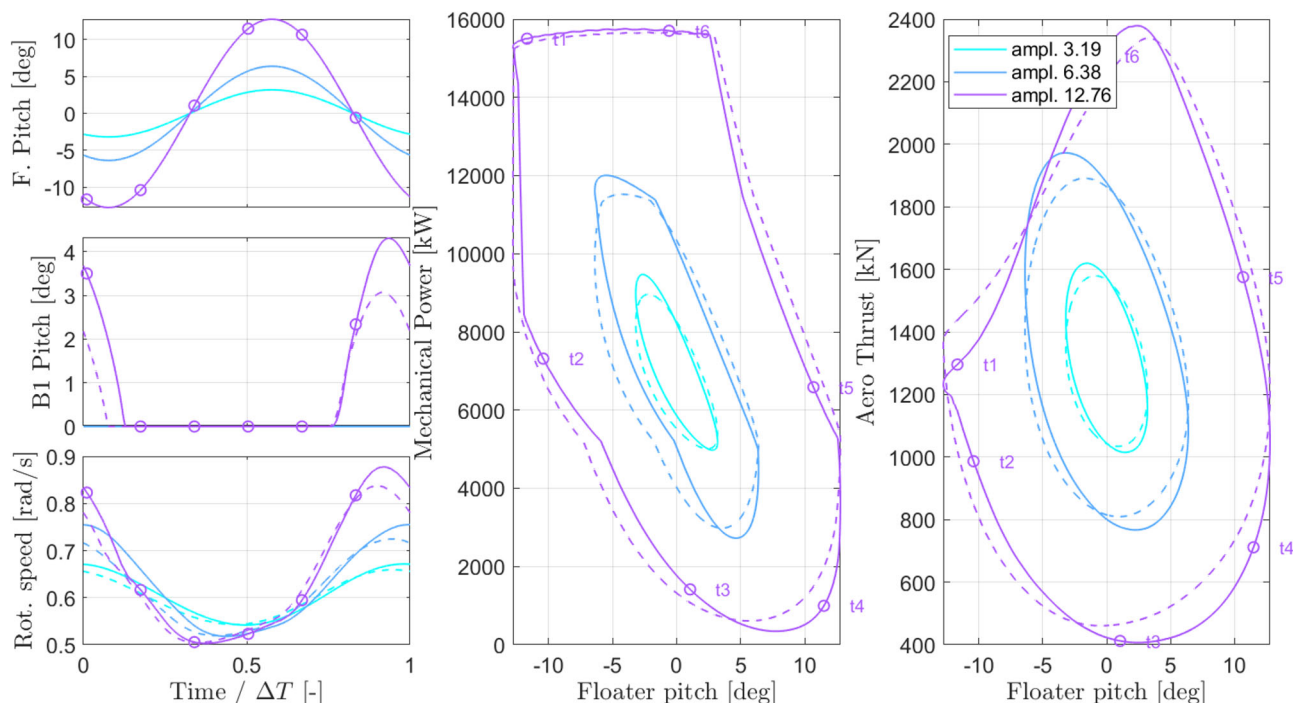


FIGURE 20 Aerodynamic quantities for the last cycle of a prescribed pitching motion of the floater with a frequency of 0.02441 Hz and different amplitudes. Wind speed of 8 m/s. Solid lines represent MIRAS-HAWC2 simulations, while dashed lines represent HAWC2-BEM

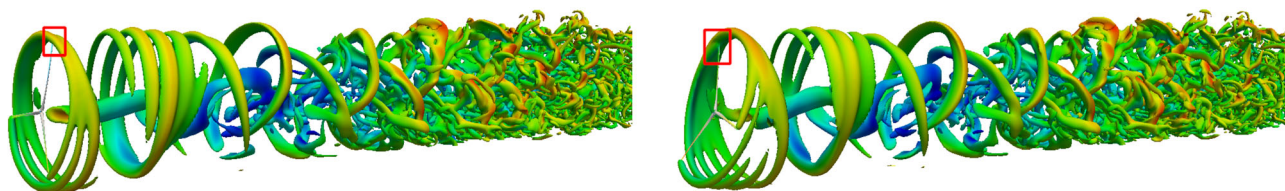


FIGURE 21 Snap-shots of three-dimensional vorticity contours over the last cycle of a floater prescribed pitch motion with amplitude 12.7 degrees and frequency 0.02441 Hz. (Left) t3 and (right) t4. Wind speed of 8 m/s

An over-speeding of the rotor has been observed for the MIRAS-HAWC2 simulations at 8 m/s, with the largest differences between the codes observed in the thrust prediction during the backwards motion of the turbine, as shown in Figure 23 where BEM under-predicts the minimum thrust by more than a 25 % while the maximum thrust is under-predicted by approximately a 7%. Differences in power are kept below 3 % throughout the complete cycle. In this case, the rotor seems to transition between different states, as it is shown in Section 6.3.

Analyzing the iso-vorticity contours, Figure 24, it can be appreciated that during most of the backwards motion of the turbine, $t3 - t4 - t5$, the rotor is operating inside its own wake, while it is generating a negative thrust force, a clear sign of a global propeller state. At the same time, large negative angles of attack can be observed from the ring distribution in Figure 25, a clear indication of negative stall in the root region between times $t4 - t5$. Moreover, a negative aerodynamic power, which is not translated into the mechanical power at the shaft, is predicted around $t3$ slowing down the rotor. It can be appreciated in the iso-vorticity contours at $t6$ that the newly generated tip vortex by the upwards pointing blade, highlighted in a red box, is released downstream of the tip vortex created by the previous passing blade. Once the turbine starts moving forward, the thrust force becomes positive but the rotor blades continue hitting the previously released vortex structures. During this first half of the forward motion, $t1 - t2$ approximately, it is observed that the airfoil sections near the blade root enter stall, as seen in Figure 25. Note that the lower and upper limits of the color-bar are truncated at -12° and 14° , respectively, these values are representative of the static stall α_{oa} in the vast majority of the airfoils that the blade mounts. It is important to emphasize here that the dynamic stall methods implemented in both the BEM and the LL solvers should be capable of quite accurately model the physics of this case. It is only in the second half of the forward motion and beginning of the transition to the backward motion, $\approx t2 - t3$, that the rotor leaves behind its own wake and appears to be in normal operation. Notice that the very long, slender, and flexible blades lag behind during the first part of the backward motion, $\approx t3 - t4$. The opposite occurs, but to a lesser extent, during the beginning of the forward motion. Unfortunately, as the rotor is immersed inside its own wake the blade

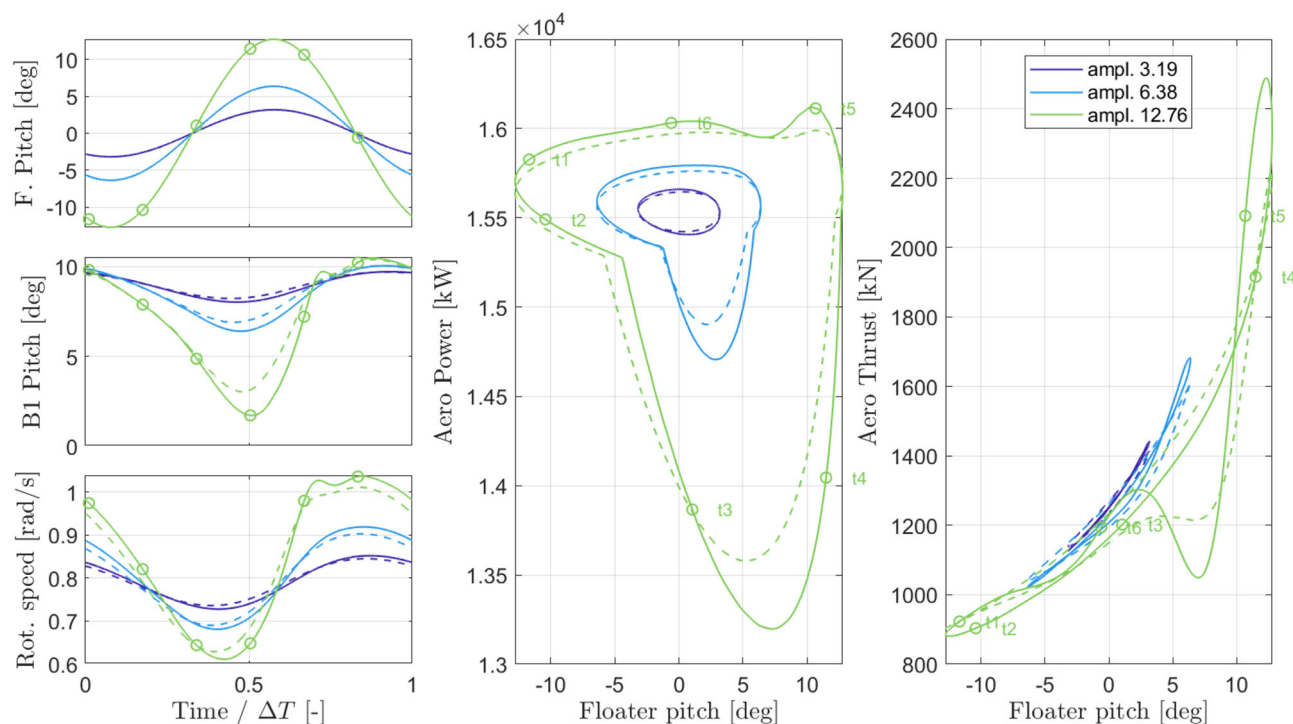


FIGURE 22 Aerodynamic quantities for the last cycle of a prescribed pitching motion of the floater with a frequency of 0.02441 Hz and different amplitudes. Wind speed of 15 m/s. Solid lines represent MIRAS-HAWC2 simulations, while dashed lines represent HAWC2-BEM

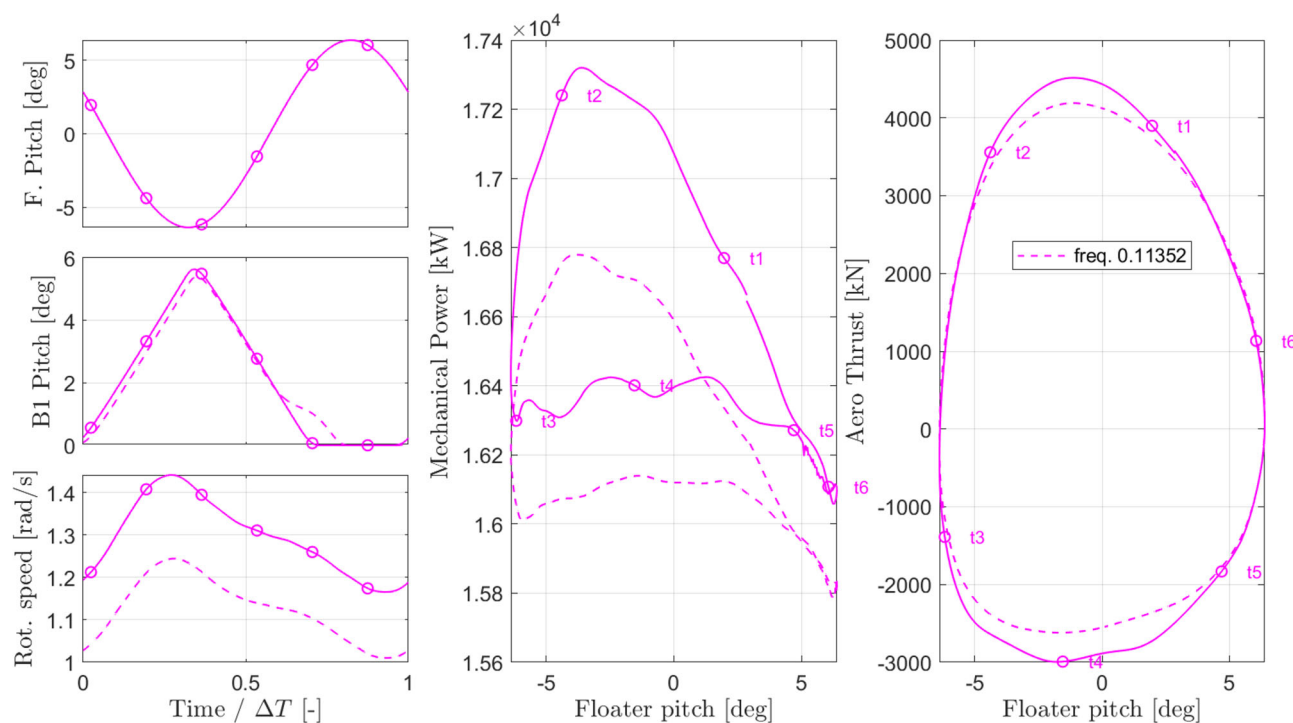


FIGURE 23 Aerodynamic quantities for the last cycle of a prescribed pitching motion of the floater with an amplitude of 6.38° and a frequency of 0.11352 Hz for a wind speed of 8 m/s. Solid lines represent MIRAS-HAWC2 simulations, while dashed lines represent HAWC2-BEM

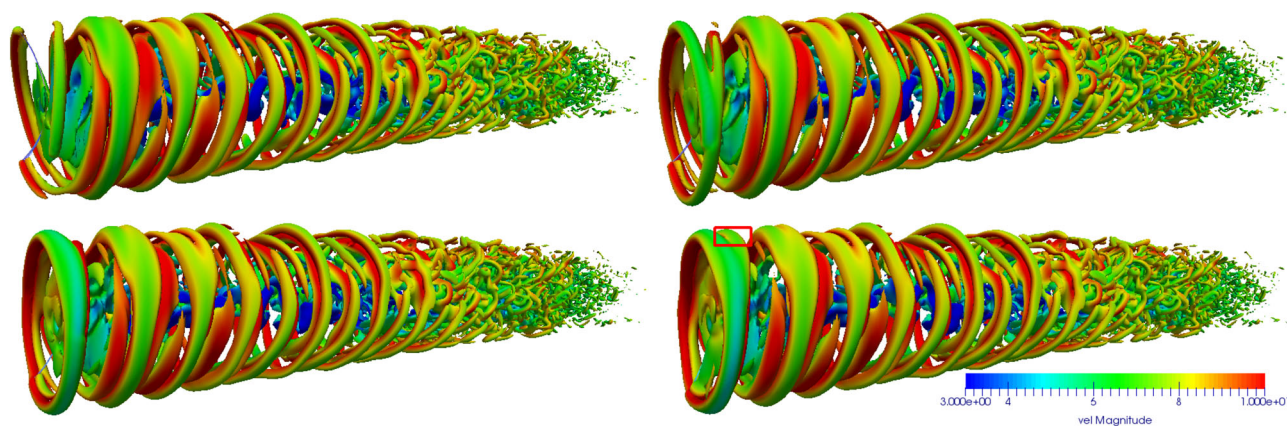


FIGURE 24 Three-dimensional iso-vorticity contours over the last cycle of a floater prescribed pitch motion with an amplitude of 6.38° and a frequency of 0.11352 Hz for a wind speed of 8 m/s. From left to right and top to bottom, t_3 , t_4 , t_5 , and t_6

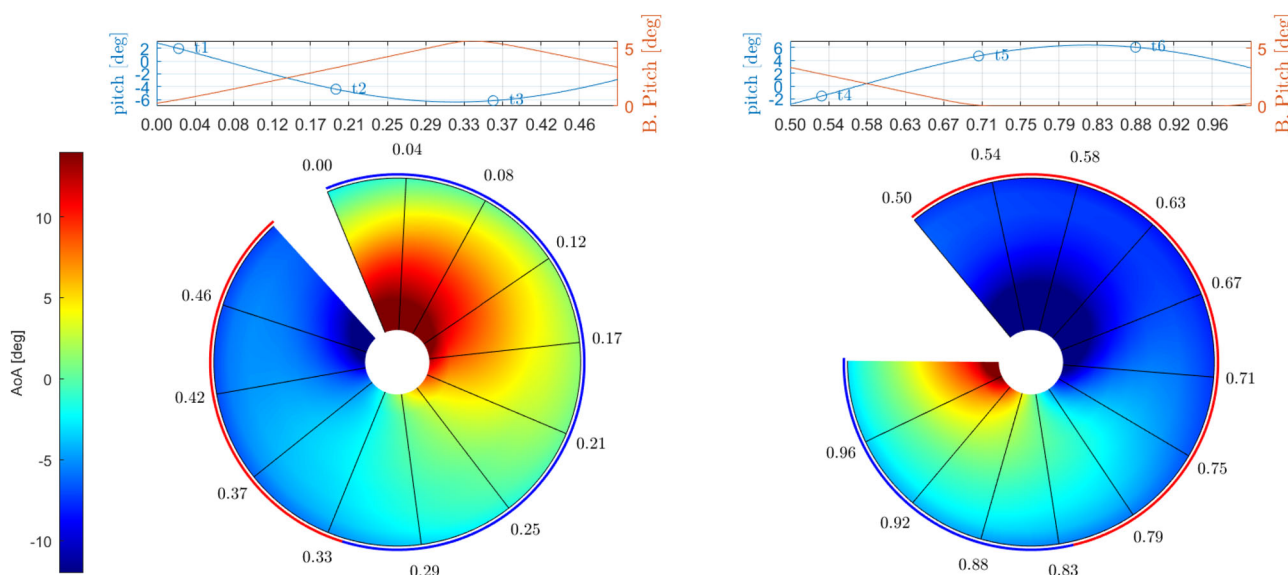


FIGURE 25 Annular distribution of the angle of attack during the last cycle of a prescribed pitch motion of the floater with an amplitude of 6.38° and a frequency of 0.11352 Hz. Wind speed of 8 m/s. Blade 1

deflections cannot be clearly seen in the iso-vorticity plots. Figure 26 is introduced to more clearly show the variation in the blade deflections in relation to the floater motion. Mark that the studied t_1 , t_2 , t_3 , t_4 , t_5 , and t_6 times are highlighted with a pink rectangle.

At 15 m/s, the rotor is over-speeding in both fidelities, pure HAWC2 and MIRAS-HAWC2. Generally, differences between the codes are much larger than observed for the 8 m/s case. Regarding the generated wake, depicted in Figure 27 for times t_2 and t_5 , it appears to be formed by much smaller turbulent structures which seem to be triggered by the quick blade deflections. Looking at the ring distributions of the angle of attack presented in Figure 28, the higher free-stream velocity has induced a larger region of the disc under positive stall while the negative stall area has been considerably reduced.

6.3 | Rotor state analysis in pitch

As seen from the previous results, the rotor operates in strongly varying inflow conditions during the pitch motion cycle. Following Lienard et al¹⁴ and Eggleston and Stoddard,⁵⁰ we now utilize the inflow velocity ratio V_c/V_h , to analyze the rotors transition between the different rotor states. V_c is the velocity normal to the rotor plane, computed as $V_c = -(V_w \cos(\theta) - \dot{\theta} l_{HT})$, mind the negative sign in accordance with helicopter notations.

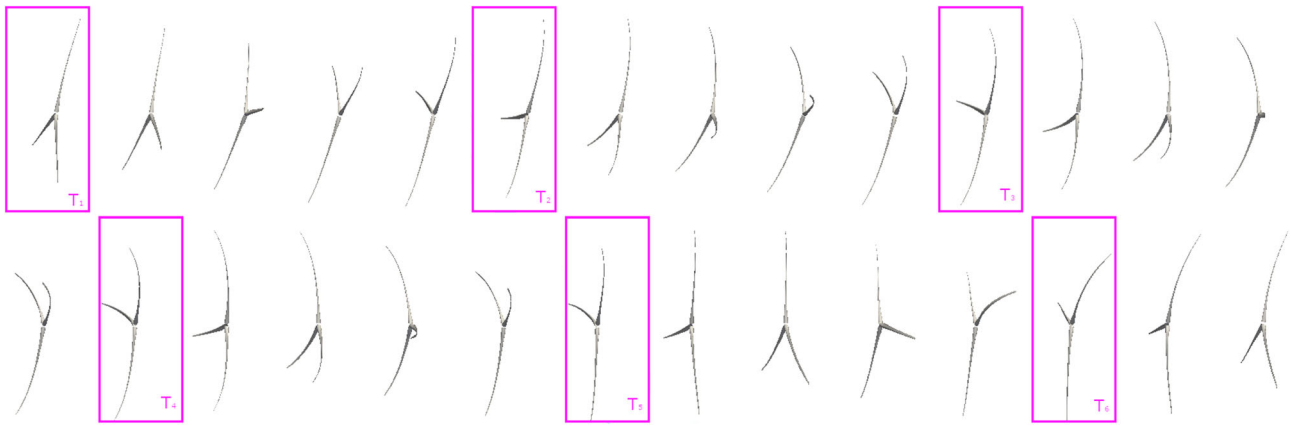


FIGURE 26 Rotor geometry for the last cycle of a prescribed pitch motion of the floater with an amplitude of 6.38° and a frequency of 0.11352 Hz. Wind speed of 8 m/s

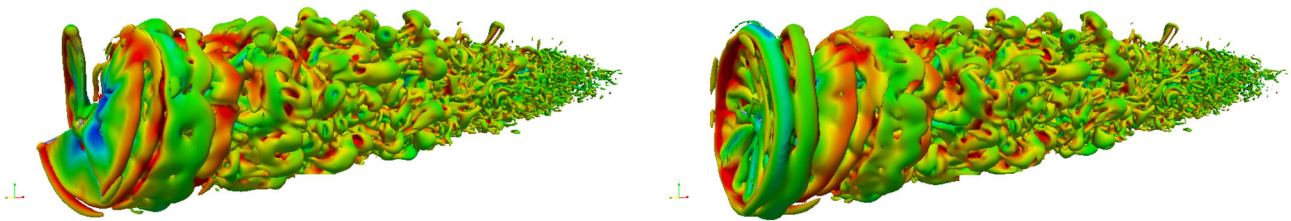


FIGURE 27 Three-dimensional iso-vorticity contours over the last cycle of an extreme floater motion at 15 m/s. With an amplitude of 6.38° and a frequency of 0.11352 Hz. (Left) t2 and (right) t5

V_w is the wind speed, and l_{HT} is the distance from the center of the turbine pitch rotation to the hub center, in this case 135 m. θ is the floater pitch angle, and $\dot{\theta}$ is the floater angular velocity. V_h is a theoretical induced velocity calculated as

$$V_h = \text{sgn}(T) \sqrt{\frac{|T|}{2\rho S}} \quad (8)$$

where T is the turbine thrust at a given time, S is the rotor area, and ρ is the air density. The use of the total turbine thrust T and the calculation of V_c at the hub height simplifies this expression, making it useful to capture trends in the turbine operation but not an accurate way to identify the different rotor states in rotatory movements of the turbine.

From actuator disc theory, a rotor can operate in different states,⁵¹ i.e., propeller state (PPS), turbine state (TBS), turbulent wake state (TWS), vortex ring state (VRS), and propeller break state (PBS). If energy is added to the fluid, the rotor normally operates in PPS, with a contracting stream-tube, and a resulting propulsion producing thrust force which is directed upstream. In TBS, the rotor extracts energy from the flow, exhibiting a stream-tube that expands downstream the rotor plane. With increasing loading of the turbine (axial induction factor), the rotor can transition to TWS, in which there is a reverse flow region at the exit of the stream-tube which progresses towards the rotor plane. The turbine is releasing very strong vortices that move very slowly downstream, violating the assumptions of inviscid and irrotational flow of the actuator disk model. When the reverse flow reaches the actuator disk, the rotor enters VRS, blocking the flow through the rotor. If the axial induction factor of the rotor is larger than unity, it can enter PBS, also known as the reverse thrust state, where power is added to the flow. But, contrary to what occurs in PPS, the thrust created is directed downstream.

The computed inflow velocity ratio over the last pitching cycle for all the prescribed pitch cases considered in this work is depicted in Figure 29. An arrow has been drawn to indicate the loop direction, which is the same for all cases. An unusual effect is observed in the simulations with a pitch frequency of 0.11352 Hz in under-rated conditions, the most extreme case. The rotor does not seem to enter the VRS transitioning from the TWS, as it has been previously reported by other researchers in simulations without active turbine control.^{14,50} It abruptly jumps from the TBS to the PPS and then smoothly transitions down to what can be called an inverted vortex ring state. The rotor exits the VRS entering again the PPS during the last part of the backward motion of the floater. When the floater transitions to forwards motion, there is again an abrupt jump down to TBS. The abrupt jumps of the inflow velocity ratio from PPS to TBS or vice-versa happen for thrust values in the vicinity of zero. The inflow velocity ratio is inversely proportional to the square root of the rotor thrust. Therefore, when the thrust is close to zero, the ratio becomes

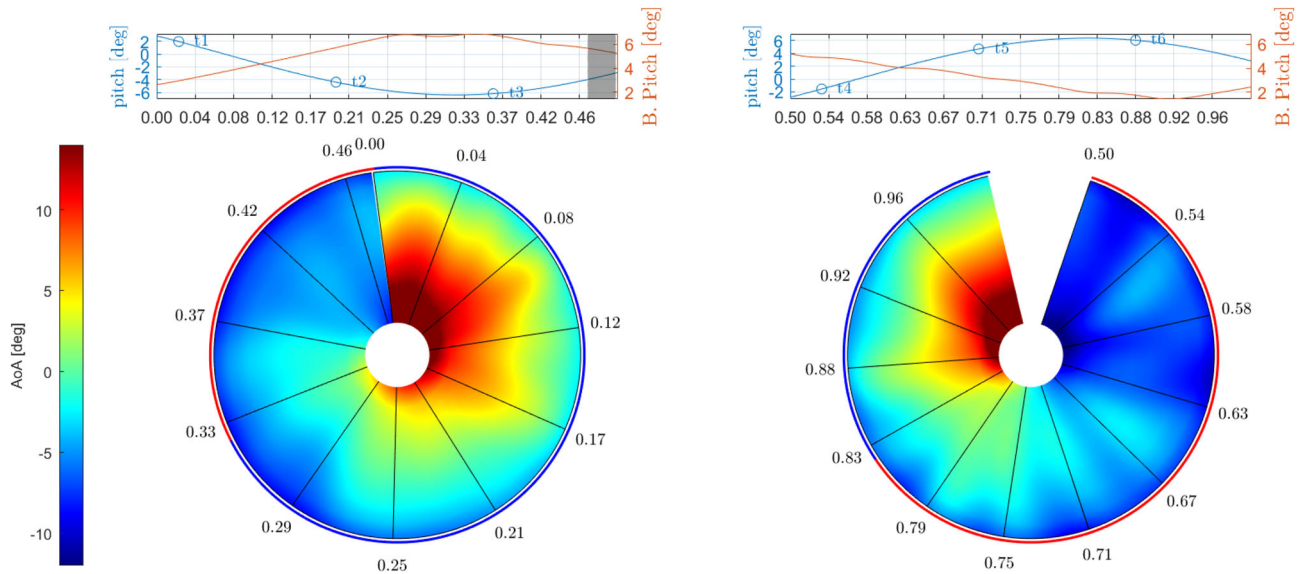


FIGURE 28 Annular distribution of the angle of attack during the last cycle of a prescribed pitch motion of the floater with an amplitude of 6.38° and a frequency of 0.11352 Hz. Wind speed of 15 m/s. Blade 1

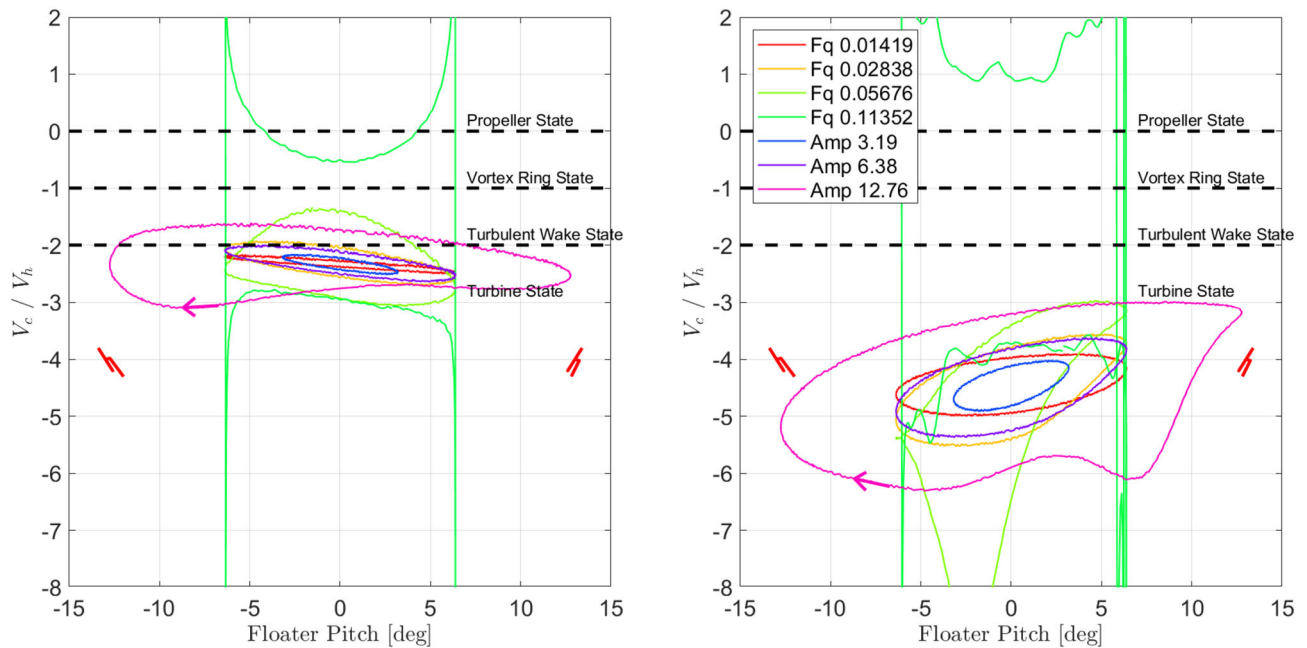


FIGURE 29 Computed inflow velocity ratio for all the harmonic pitch cases with active turbine control (left) 8 m/s and (right) 15 m/s cases

a large absolute number. Note that, for the rest of the considered cases, the rotor seems to be far from entering a global VRS. In many cases, it is operating in TWS during part of the backwards motion, and as previously shown during the analysis of the iso-vorticity contours, the top part of the rotor can enter local VRS. At 15 m/s, the rotor operates in normal turbine state in all cases but for the extreme frequency, where it operates in PPS during the full backwards motion. The spikes observed during the transition to forward motion are related to the blade instabilities that make the turbine thrust jump from positive to negative values and vice-versa multiple times.

To further investigate this phenomena, a set of cases where the rotor is close to enter VRS have been simulated with the controller turned off. The wind speed of 8 m/s has been selected since it has been found that the rotor is more prone to enter non-conventional operation states at lower wind speeds. The rotational speed was fixed to 0.6025 rad/s, and the blade pitch was set to 0° , following the turbine normal operation

points. As can be seen in Figure 30, for a pitching frequency of 0.07804 Hz, the rotor transitions smoothly between the states TBS-TWS-VRS. However, if the frequency is higher, the rotor enters PPS, and it transitions abruptly down to TBS as it has been previously shown. This can happen multiple times during the backwards motion of the floater, as seen for the 0.11352 Hz case, in which the wind speed is lower than the velocity generated by the motion of the turbine, i.e., positive V_c , and if the thrust is negative, i.e., negative V_h , the inflow velocity ratio becomes negative but the relative wind seen by the rotor is coming from downstream. Therefore, it can be said that the turbine is operating in an inverted fashion, in which it can transition again between all states, as seen during part of the backwards motion for the 0.11352 Hz case.

Note that the flexibility of the blades and tower should add an extra velocity term to the inflow velocity ratio calculation, which is not taken into account in our analysis for the sake of simplicity.

6.4 | Maximum blade and tower loading in pitch

The thrust variations associated with the pitch motion naturally alters the structural loads. Bar diagrams with the maximum attained blade root and tower base bending moments, BRBM and TBBM, respectively, are shown in Figure 31. A comparison between the MIRAS-HAWC2 and HACW2-BEM predictions shows that for most of the cases the higher fidelity model predicts just slightly larger maximum values of the blade root bending moments. However, for the extreme cases, i.e., pitching frequency of 0.11352 Hz, the difference is much larger, especially in above-rated conditions. Such large differences are accentuated in the tower base bending moments, mostly for the side-side component. Note that, as an exception, the BEM method generally over-predicts the fore-aft tower base bending moment, except for the extreme case. It is observed that increasing the frequency or the amplitude of the prescribed floater motion raises the maximum moments. The only exception is the maximum torsional tower bending moment at the wind speed of 15 m/s, which seems to be almost invariant to frequency or amplitude changes (disregarding the extreme frequency case). Moreover, under-rated wind conditions exhibit the larger maximum blade root bending moments, except for the extreme case and the LL prediction of the flap-wise component in the 0.05676 Hz case.

6.5 | Wake analysis in pitch

The influence of the turbine motion in the downstream development of the wake is studied in what follows. First, the integrated wake velocity recovery, Equation 6, and the integrated turbulence intensity, Equation 7, are plotted as function of the downstream distance to the rotor plane in Figure 32. In this case, a prescribed harmonic pitch motion with 6.38° amplitude and three different frequencies are compared to the

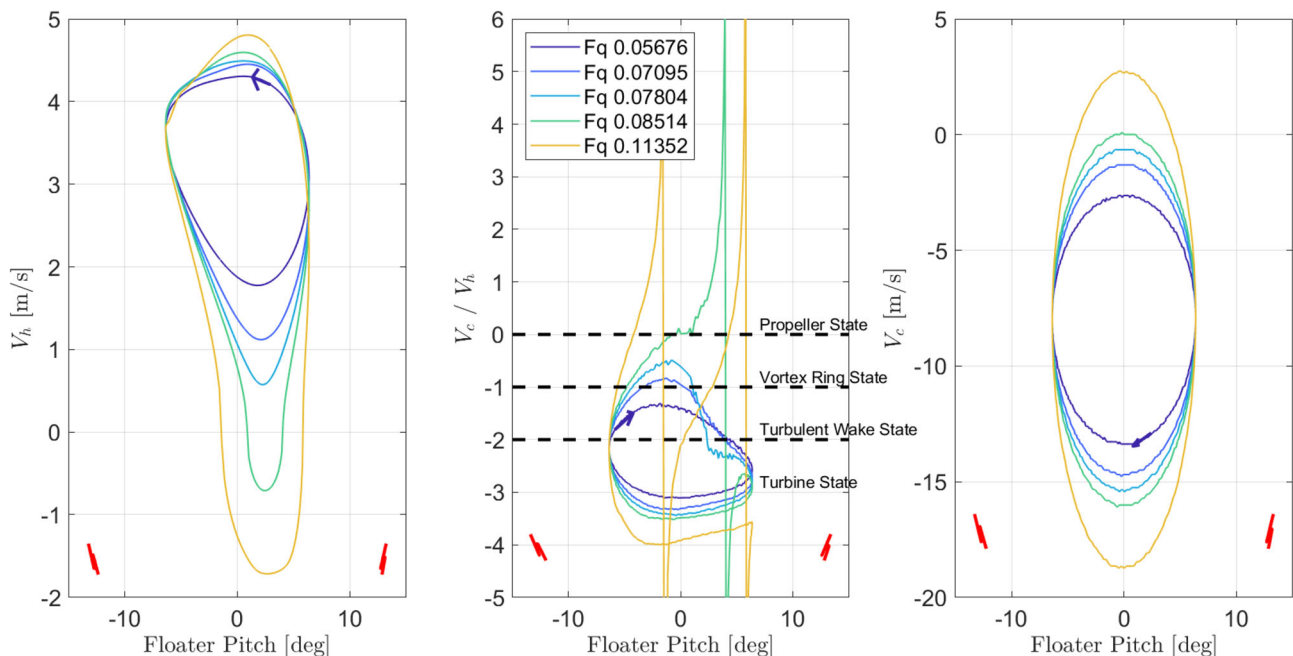


FIGURE 30 (Left) theoretical induced velocity, V_h , (middle) inflow velocity ratio, V_c/V_h , and (right) normal velocity to the rotor plane V_c for selected harmonic pitch cases with the controller turned off. Wind speed of 8 m/s

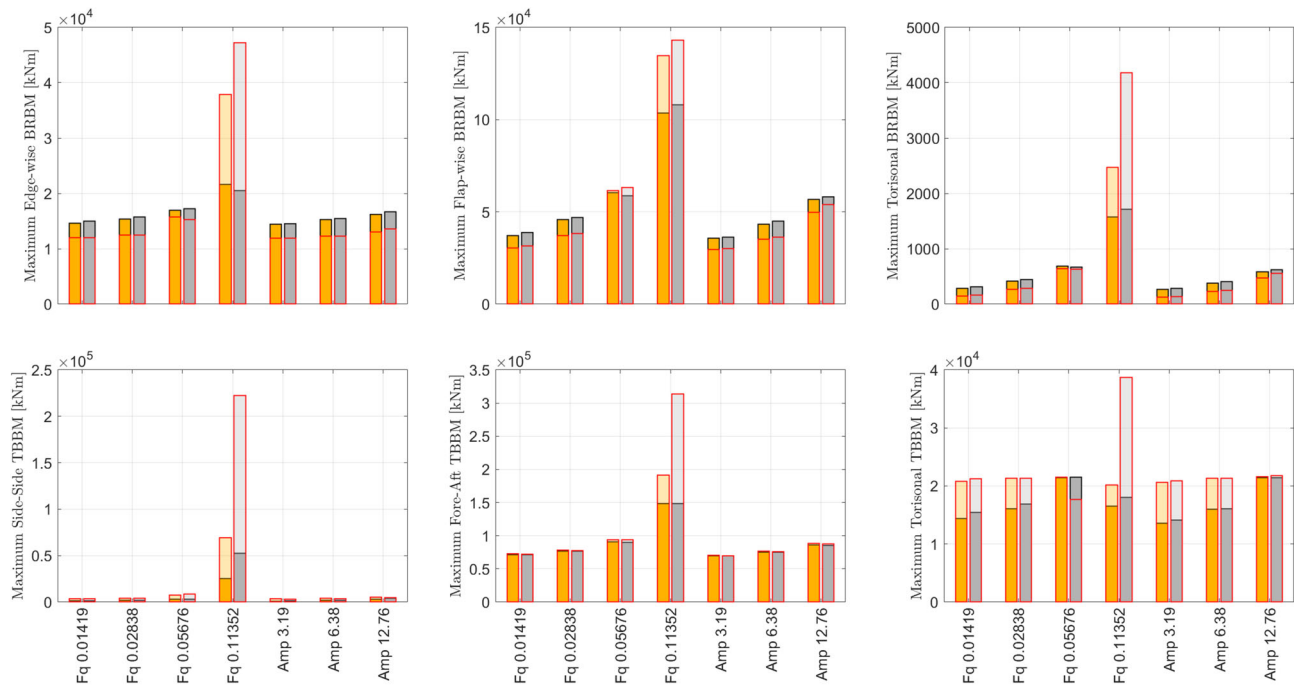


FIGURE 31 Maximum computed (top) blade root and (bottom) tower base bending moments for the prescribed pitching motion of the floater at wind speeds of 8 m/s, black edged bars, and 15 m/s, red edged bars. Bars with amber filling represent HAWC2-BEM, while bars with gray filling represent MIRAS-HAWC2 simulations

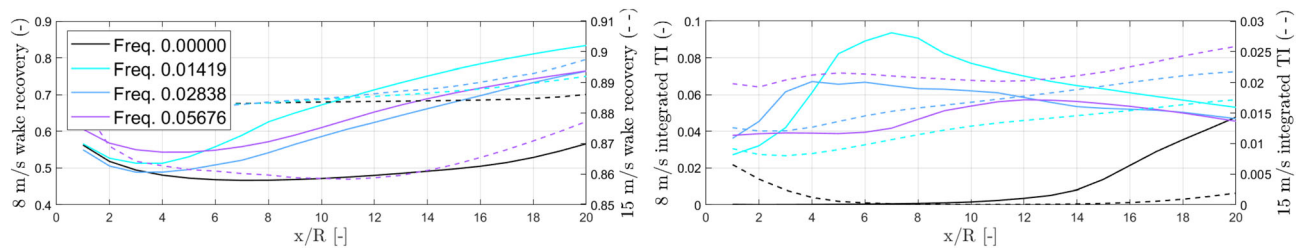


FIGURE 32 Wake analysis for a prescribed pitching motion of the floater with an amplitude of 6.38° and different frequencies at wind speeds of 8 m/s (solid lines) and 15 m/s (dashed lines). (Left) Wake velocity recovery; (right) Integrated Turbulent Intensity

non-oscillating case (bottom-fixed turbine), referred as Freq. 0.00000. Cases at a wind speed of 8 m/s are depicted in solid lines, while dashed lines represent the cases at 15 m/s. In below-rated conditions, there is a large difference in the wake recovery of the three pitching cases compared to the non-oscillating case, which shows a much slower recovery far downstream and a lower integrated TI. This is in agreement with the experimental observations of Fu et al,²² who observed a faster wake recovery of a pitching turbine due to an enhanced turbulent mixing that led to a higher turbulence kinetic energy level. Initially, the higher frequency case (0.05676 Hz) shows a better initial recovery compared to the other cases. However, there is a change in tendency and the lowest frequency motion (0.01419 Hz) reaches the best recovery far downstream, approximately 50% that of the bottom-fixed case. Regarding above-rated conditions, the highest frequency case shows the worst wake recovery independently of the downstream location, even worse than the non-oscillating case. This is surprising since the turbulence intensity level achieved is larger than the rest. It was initially assumed that more mixing would accelerate the recovery but this does not seem to hold for this case. The other three cases, including the non-oscillating case, show very similar characteristics in the near wake. Differences grow downstream of the 10R location, with the 0.02838 Hz case displaying the best recovery at the 20R location. Figure 33 shows the integral wake velocity and turbulence intensity for three different amplitudes of the pitch oscillation, at the given frequency of 0.02441 Hz. It is clearly observed that the wake recovery far downstream improves with the increasing amplitude, with the 12.7° case attaining the best recovery 20 R downstream in both below- and above-rated conditions and with an increase from 0.48 to 0.63 in wake recovery at $x/R = 12$ for the motion at 0.02838 Hz at 8 m/s.

In order to assess the influence of large blade motions (including floater oscillation and the blade elastic deformation) in the recovery of the wake, we have investigated the blade fore-aft tip displacement for the cases with a prescribed harmonic pitch motion of the floater. Focusing first

in below rated wind conditions, Figure 34 to the left shows the fore-aft displacement of the tip of blade one over 350 s of simulated time. The peaks where the displacement is larger than 80% of the largest displacement are located and marked in the figure. Large displacements are linked to substantial variations in the blade loading, the wake geometry and the strength of the released tip vortex and can promote a faster break down of the wake structure. Moreover, Figure 34 to the right presents the time interval between consecutive peaks. Thus, each filled up circle represents a peak pair, and it is colored by the mean value of its fore-aft displacement. Note that 1,275 s of simulated time have been used for the peak investigation; the first 125 s have been disregarded to avoid transients. It is observed from the sub-figure to the right that the 0.01419 Hz case has the largest amount of peak pairs with the shortest time between them, approximately 10 s. Followed by the high frequency case, i.e., 0.05676 Hz, with a time interval of approximately 19 s. Finally, in the 0.02441 Hz case, all the peaks appear much more spaced in time, with an interval between 30 and 40 s. A low time interval between high consecutive peaks is an indication of a reduced distance between consecutive tip vortices, which could promote a vortex instability and trigger a faster wake recovery. Our findings are therefore in agreement with the wake velocity recovery results presented with solid lines in Figure 32, left, where the lowest frequency pitch oscillation achieves the fastest wake recovery, followed by the highest frequency oscillation and with the mid-range oscillation presenting the slowest recovery of the three cases. To further support our analysis, we have carried out a similar peak study in above rated conditions, i.e., 15 m/s. Note that in this case, the highest frequency floater oscillation showed surprisingly the worst wake recovery, earlier presented in Figure 32 to the left using dashed lines. This finding is in agreement with the peak analysis depicted in Figure 35 where a minimum peak to peak time interval of approximately 7.5 s is computed for the 0.01419 and 0.02441 Hz cases and almost a three times larger time gap between large tip deflections is recurrently observed for the highest frequency floater oscillation.

7 | INVESTIGATION OF THE EFFECT OF FLOATER HARMONIC MOTION IN SURGE

The test matrix for the harmonic surge motion is presented in Table 4. In general, similar results are observed during the isolated surge translation compared to the pitch rotation; therefore, only selected results are presented in what follows.

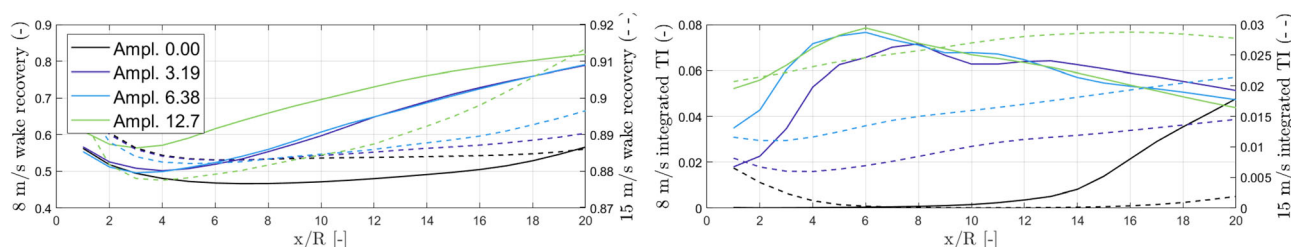


FIGURE 33 Wake analysis for a prescribed pitching motion of the floater with an frequency of 0.02441 Hz and different amplitudes at wind speeds of 8 m/s (solid lines) and 15 m/s (dashed lines). (Left) Wake velocity recovery; (right) Integrated Turbulent Intensity

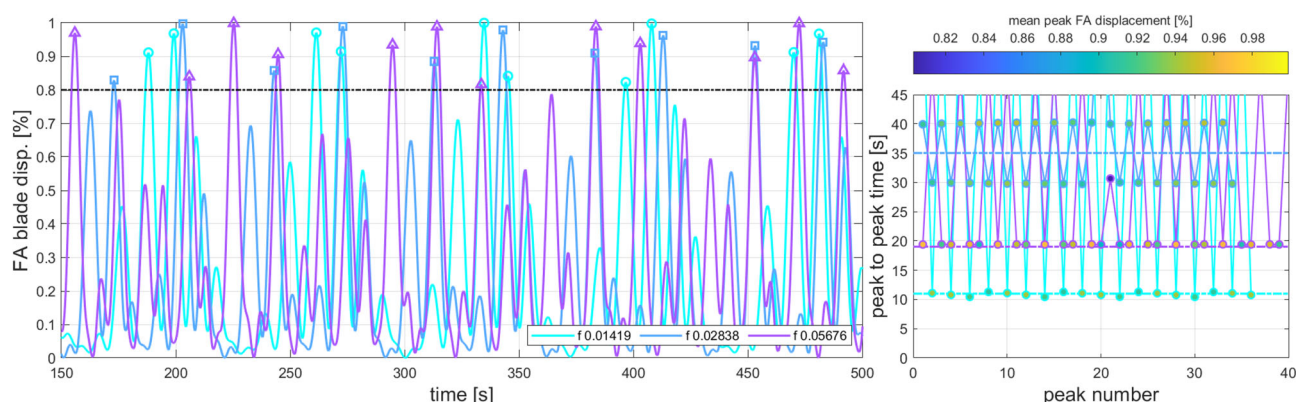


FIGURE 34 Peak analysis of the blade one tip position for a prescribed pitching motion of the floater with an amplitude of 6.38° and different frequencies at a wind speed of 8 m/s

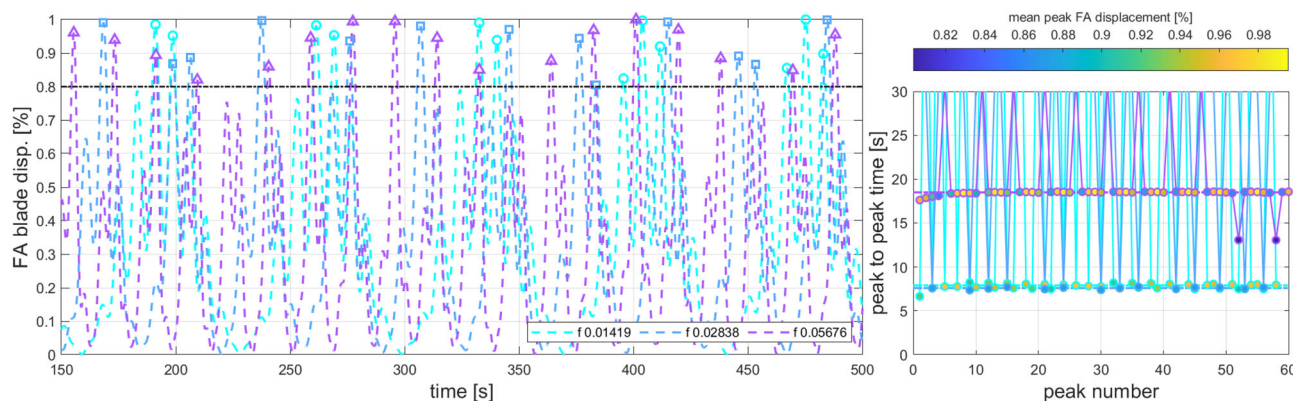


FIGURE 35 Peak analysis of the blade one tip position for a prescribed pitching motion of the floater with an amplitude of 6.38° and different frequencies at a wind speed of 15 m/s

7.1 | Turbine behavior and iso-vorticity comparison in surge

The influence of the surge frequency in the turbine behavior is herein presented for a wind speed of 8 m/s and an amplitude of 15 m. The key sensors presented in Figure 39a show an almost identical picture as the pitch study in below-rated conditions. Even so, the wake has a very different structure due to the transitory nature of the movement, as seen in Figure 36. The iso-vorticity snap-shot at t_1 shows the wake during the floater backward motion, while at t_5 it is depicted during the forward motion. In the latter one, a wider spacing between the newly released tip vortices and the ones released by the previous passing blade is observed. Merging of tip vortices during part of the backwards motion is observed, with the thickness of the vortices facilitating the merging although there are no clear signs of direct blade-vortex interaction. As it is shown in Section 7.3, the turbine operates in turbulent wake state during most of the rearward movement.

Results for a surge motion with an amplitude of 15 m and frequencies of 0.01419, 0.2838, and 0.05676 Hz in above-rated conditions are presented below. The change in selected rotor quantities over the last cycle is summarized in Figure 39b. A fairly good agreement between the solvers is obtained at the lower frequencies; however, large discrepancies are seen at 0.05676 Hz. In this case the width of the thrust loop predicted by LL is narrower than for the equivalent pitch case. And generally, differences between the BEM and LL simulations are much larger. Notice how BEM under-predicts both the maximum and the minimum power and thrust compared to the LL solution. Iso-vorticity contours for two selected times, i.e., t_1 and t_4 , are reproduced in Figure 37.

Selected sensors for the amplitude surge study at the frequency of 0.01419 Hz and a wind speed of 8 m/s are presented in Figure 39c. The largest differences in absolute terms between the codes are observed during the forward motion of the blade, $t_1 - t_2 - t_3$. Again, such differences grow with the increasing amplitude of the surge. In general, smaller differences are appreciated during the backward motion, $t_4 - t_5 - t_6$, although these differences are larger in relative terms. The BEM method consistently under-predicts the height of the power and thrust loops by up to a 20% and 17%, respectively, for the highest amplitude case. Three-dimensional iso-vorticity contours are shown in Figure 38 for the selected times t_2 and t_5 . A very distinctive wake shape is simulated in this case. The tip vortex structures seem to dissipate faster and do not reach the far wake. An accordion effect is clearly observed when comparing the wake during the forward, t_2 , and backward, t_5 , motion of the floater.

The final study regarding the surge DoF focuses on an amplitude variation in above-rated conditions. Surge amplitudes of 5, 10, and 15 m are considered, and key outputs are presented in Figure 39d. Again in this case, the LL solver predicts a larger loop height compared to the BEM predictions. With differences up to 10% in power and 15% in thrust. As expected, the differences in power between BEM and LL predictions are reduced compared to the below-rated case. The iso-vorticity contours for the 15 m amplitude case at times t_1 and t_5 are shown in Figure 40. The tip vortex is very stable compared to the previous cases in below-rated conditions, remaining unaltered quite far downstream. It is surprising that even the root vortex remains almost unaltered until the end of the computational domain.

7.2 | Turbine behavior and iso-vorticity comparison for extreme cases in surge

A very similar behavior of the rotor is observed for the below-rated case with a surge frequency of 0.11352 Hz as for the equivalent pitch case and therefore the results are not included for the sake of brevity.

At 15 m/s, the surging rotor presents a very abrupt flap-wise instability at 1.45 Hz, probably flutter. And as shown in Figure 41, large blade deformations generate a more chaotic wake than the one observed in the equivalent pitch case (Figure 27). This large discrepancies with the

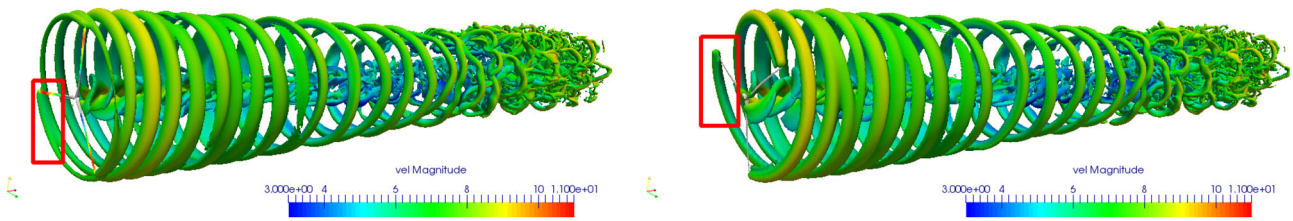


FIGURE 36 Three-dimensional iso-vorticity contours over the last cycle of a floater prescribed surge motion with an amplitude of 15 m and a frequency of 0.05676 Hz. (Left) t1 and (right) t5. Wind speed of 8 m/s

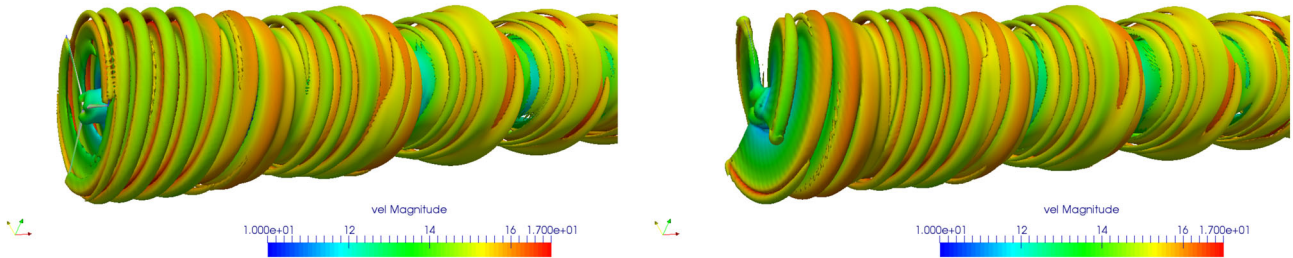


FIGURE 37 Three-dimensional iso-vorticity contours over the last cycle of a floater prescribed surge motion with an amplitude of 15 m and a frequency of 0.05676 Hz. (left) t1 and (right) t4. Wind speed of 15 m/s

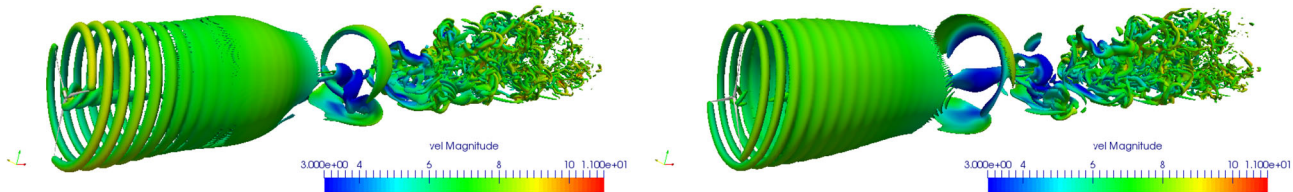


FIGURE 38 Three-dimensional iso-vorticity contours over the last cycle of a floater prescribed surge motion with an amplitude of 15 m and a frequency of 0.01419 Hz. (Left) t2 and (right) t5. Wind speed of 8 m/s

floaters motion in pitch can clearly be appreciated in the angle of attack ring distributions presented in Figure 42, where on top of the positive stall region in the vicinity of the root (already seen in the pitch cases), an extreme behavior is observed in the tip region during most of the backward motion where the outer part of the blade enters and exits negative stall at a very high frequency rate. Fairly similar loads are predicted by both fidelity codes although it is very difficult to assess if this very challenging scenario is correctly captured by the current levels of fidelity. The codes can probably predict the onset of the aeroelastic instability, but what is very challenging is the prediction of its amplitude.

7.3 | Rotor state analysis in surge

The variation of inflow velocity ratio, V_c/V_{tr} , over the last surge cycle is presented in Figure 43. In surge, V_c is computed as $V_c = -(V_w - \dot{x})$, where x is the prescribed surge position. Since surge is a translatory motion, in the present analysis, it is assumed that the full rotor is moving with the same velocity. Therefore, the inflow velocity ratio is a more accurate way to identify the different rotor states than during a pitch rotation. Focusing first on the 0.05676 Hz case in below-rated conditions, it is hereby confirmed that the rotor does not enter VRS, remaining in TWS during most of the backward motion. Regarding the extreme case, frequency of 0.11352 Hz, the general behavior of the rotor is very similar to the equivalent pitch case, and it will not be further discussed. At a wind speed of 15 m/s, cases with a frequency of 0.05676 Hz and lower exhibit a rotor operating in turbine state. For the 0.11352 Hz case, the rotor is operating in PPS during most of the backward motion of the turbine with the large blade deflections triggering sharp transitions between TBS and PPS.

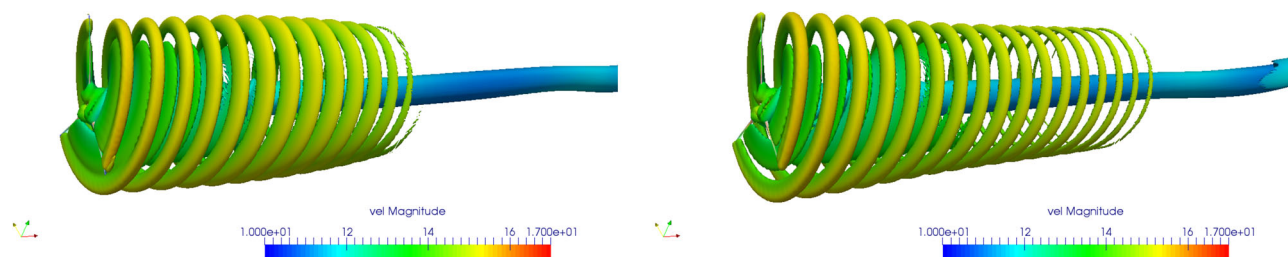
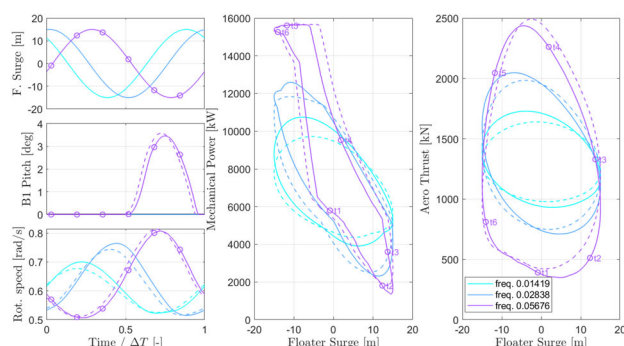
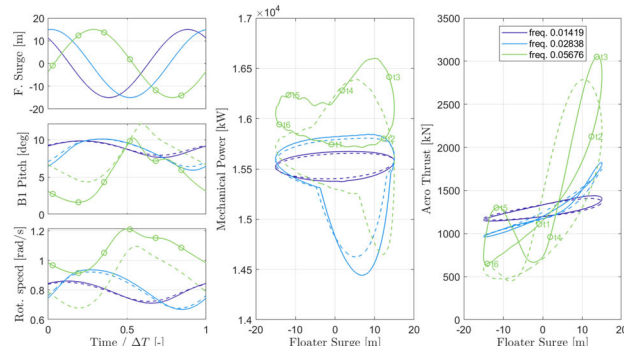


FIGURE 39 Aerodynamic quantities for the last cycle of a prescribed surge motion of the floater. Solid lines represent MIRAS-HAWC2 simulations while dashed lines represent HAWC2-BEM. (a) Amplitude of 15 m and different frequencies. Wind speed of 8 m/s. (b) Amplitude of 15 m and different frequencies. Wind speed of 15 m/s. (c) Frequency of 0.01419 Hz and different amplitudes. Wind speed of 8 m/s. (d) Frequency of 0.01419 Hz and different amplitudes. Wind speed of 15 m/s

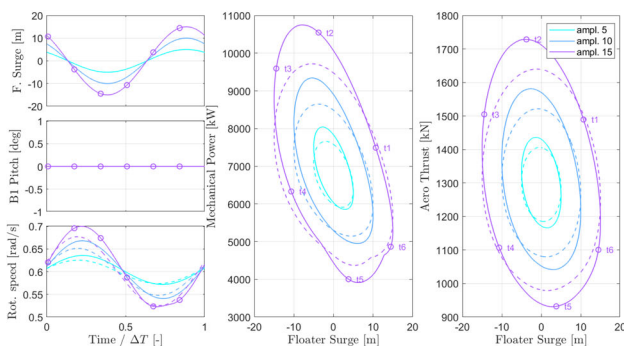


(A) Amplitude of 15 meters

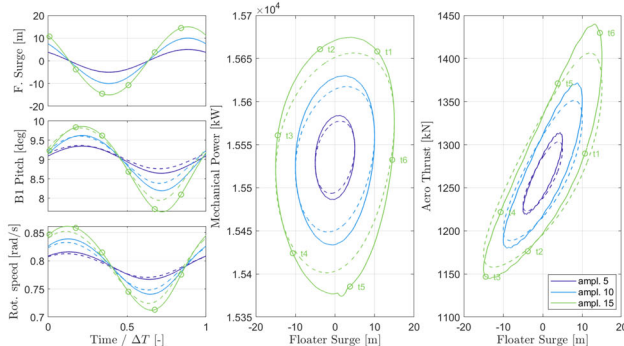
encies. Wind speed of 8 m/s.



(B) Amplitude of 15 meters and different frequencies. Wind speed of 15 m/s.



(C) Frequency of 0.01419 Hz and different amplitudes. Wind speed of 8 m/s.



(D) Frequency of 0.01419 Hz and different amplitudes. Wind speed of 15 m/s.

FIGURE 40 Three-dimensional iso-vorticity contours over the last cycle of a floater prescribed surge motion with and amplitude of 15 m and a frequency of 0.01419 Hz. (Left) t_1 and (right) t_5 . Wind speed of 15 m/s

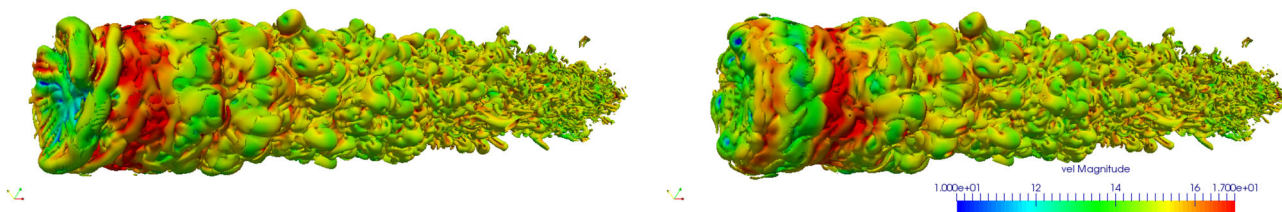


FIGURE 41 Three-dimensional vorticity contours over the last cycle of an extreme floater prescribed surge motion at 15 m/s. Amplitude of 15 m and a frequency of 0.11352 Hz. (Left) t_3 and (right) t_6

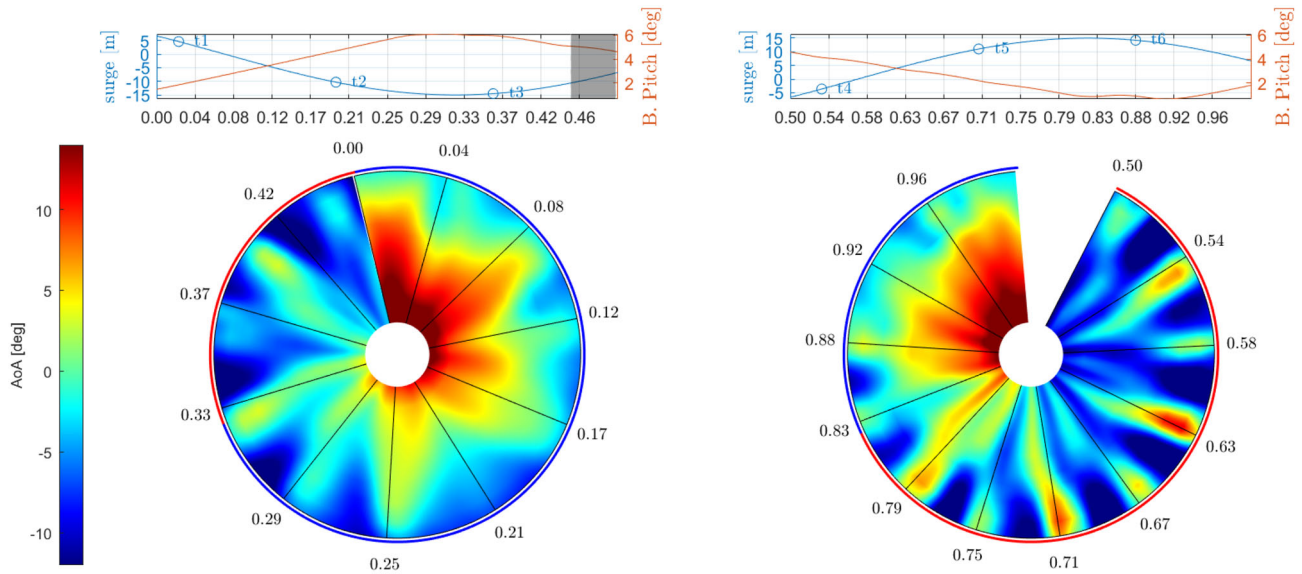


FIGURE 42 Annular distribution of the angle of attack during the last cycle of a prescribed surge motion of the floater with an amplitude of 15 m and a frequency of 0.11352 Hz. Wind speed of 15 m/s. Blade 1

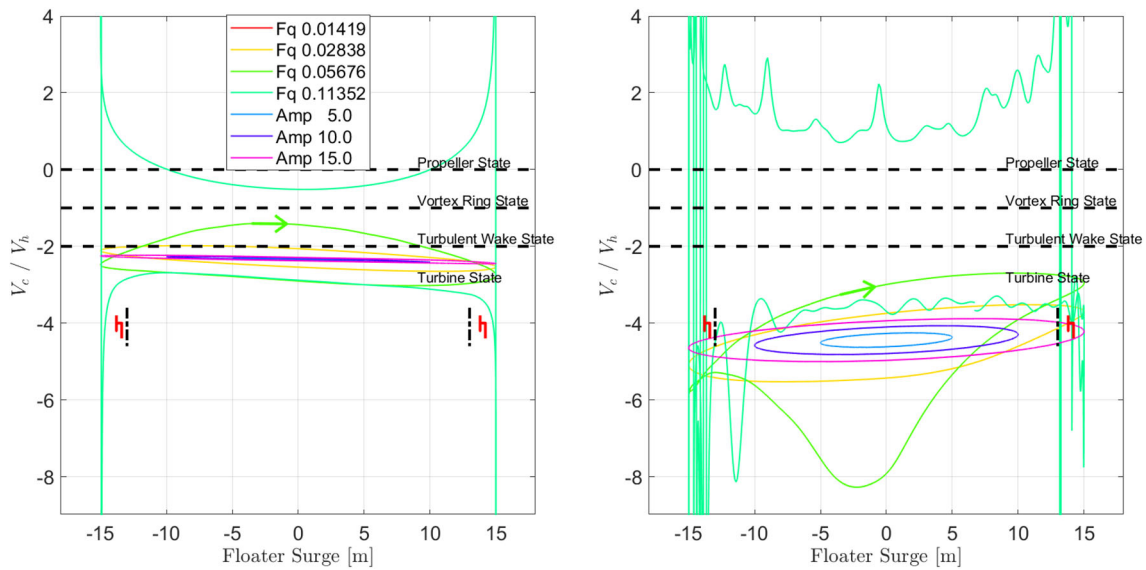


FIGURE 43 Computed inflow velocity ratio for the harmonic surge cases: (left) 8 m/s and (right) 15 m/s cases

7.4 | Maximum blade and tower loads in surge

The analysis of maximum blade root and tower base bending moments showed very similar trends as for the pitch DoF, previously shown in Figure 31. Therefore, the results are not included in the interest of brevity.

7.5 | Wake analysis in surge

Regarding the influence of the surge frequency in the recovery of the wake, our investigation shows a very similar picture as in the pitch case with small variations as seen in Figure 44. At $x/R=12$ for the motion at 0.01419 Hz, the wake recovery increases from 0.48 to 0.71 due to the surge motion in the 8 m/s case. The main differences to the pitch case are observed in the amplitude investigation, depicted in Figure 45 for a

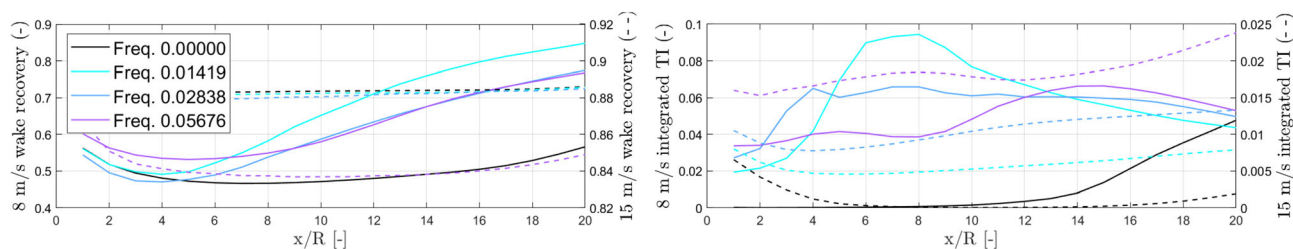


FIGURE 44 Wake analysis for a prescribed surge motion of the floater with an amplitude of 15 m and different frequencies at wind speeds of 8 m/s (solid lines) and 15 m/s (dashed lines). (Left) Wake velocity recovery; (right) Integrated Turbulent Intensity

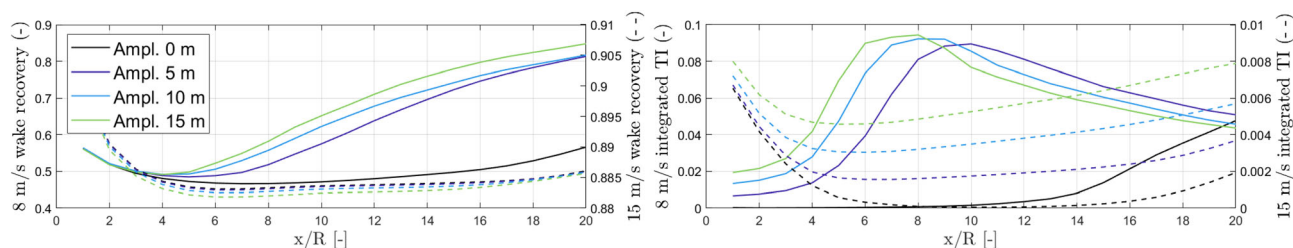


FIGURE 45 Wake analysis for a prescribed surge motion of the floater with an frequency of 0.02441 Hz and different amplitudes at wind speeds of 8 m/s (solid lines) and 15 m/s (dashed lines). (Top) Wake velocity recovery; (bottom) Integrated Turbulent Intensity

prescribed surge motion with amplitudes of 5, 10, and 15 m at a frequency of 0.01419 Hz, in the vicinity of the natural frequency of the WindCrete floater. Here, we can observe larger differences than for the pitch case in above-rated conditions, i.e., 15 m/s. In the surge case, the amplitude of the motion does not seem to have as strong effect in the recovery of the wake although the TI level is clearly affected by it, contrary to what has earlier been observed during the pitch study.

8 | CONCLUSION

A detailed investigation of the floating IEA Wind 15 MW reference wind turbine and its wake has been carried out under a large variety of conditions. The main numerical approach relied on the in-house multi-fidelity vortex solver MIRAS, equipped with a LL aerodynamic model and coupled with the multi-body finite-element solver HAWC2. The investigation was divided in four parts: (i) simulations of the turbine mounted on the WindCrete floater; (ii) study of the effect of mean tilt in the wake development; (iii) parametric study under prescribed platform pitch; and (iv) under prescribed surge motion of the assembly. Throughout the investigation, the turbine behavior predicted by the MIRAS-HAWC2 coupling has been compared against the HAWC2-BEM lower fidelity method. In HAWC2-BEM, the aerodynamic loads were estimated with the BEM approach, which is considered an industry standard, while the same structural model was kept. The systematic comparison of MIRAS-HAWC2 and HAWC2-BEM approaches allow to assess the influence of the accuracy of the aerodynamic modeling under a large variety of floating conditions.

The aero-hydro-servo-elastic simulations showed the largest differences between the codes for high wind speeds under regular waves. Regarding the influence of the wave characteristics, in general, it has been found that differences between BEM and LL methods are more sensitive to variations in wave frequency than in wave amplitude, with the BEM predictions deviating considerably from the LL solution at high wave frequencies, where BEM over-predicts the maximum amplitude of the motion by more than a 50%.

The analysis of the effect of mean tilt in the wake development has confirmed that a positive tilt angle deflects the wake upwards, promoting the wake stretching and helping break down the cylindrical wake shape, which leads to a faster wake recovery. The recovery has been quantified as function of the downwind position for three tilt angles. For a tilt angle of 10° , the recovery is 0.52 at a distance of 12 rotor radii, relative to a value of 0.45 for the 0° case.

During the parametric study under prescribed pitch motion, it was found that the largest relative differences between the codes appear during the backward motion of the rotor, where the BEM method can violate the momentum balance assumption. The largest absolute differences between the codes have generally been found at the maximum thrust and power values during the forward motion of the turbine. Regarding the

rotor flow, it has been observed that the outer part of the blades are more prone to local vortex ring state in below-rated conditions and high frequency motions. Wake recovery is enhanced by the pitch motion, since the rotor motion accelerates the mixing and breakdown of the tip vortices. A relation between short time intervals of large FA blade tip displacements and a faster break down of the wake has been demonstrated. For the case of 0.02838 Hz pitch motion at 6.38° amplitude, the wake recovery at a 12 R distance downstream increased from 0.48 to 0.63 at 8 m/s. It was further found that in terms of the wake recovery, slow oscillations have shown to be more beneficial than high frequency ones. The latter ones can produce a stronger and more stable wake, which recovers slower than in the steady case without motion. This effect is more pronounced in the surge motion, in which the delay is maintained in the far wake, contrary to what was observed for the pitch case. It has been shown that a pitching wind turbine can transition throughout all possible rotor states. Moreover, inverted rotor states have been identified and defined during a cycle with high frequency pitch motion. In this regime, negative thrust and strong blade oscillations can occur during the motion cycle. While the structural loads generally compared quite well between the BEM and vortex code results, this case showed substantial differences.

Regarding the surge investigation, it has been found that a surging rotor has, in general, a very similar behavior to a pitching one. Surge motion of 15 m amplitude at 0.01419 Hz increased the wake recovery from 0.48 to 0.71 at $x = 12R$ and 8 m/s. However, there are clear differences in the generated wake due to the translatory nature of the surge movement. The wake of a surging turbine seems to be less prone to local vortex ring state compared to the wake of a pitching turbine. Additionally, an accordion-like effect has been clearly identified in some of the presented cases. Regarding the wake recovery, the amplitude of the surge motion does not have a strong effect in the recovery of the wake for the investigated frequencies.

Overall, the study demonstrates the applicability of vortex solvers to supplement and detail the Engineering calculations offered by BEM methods for the loads and energy yield of floating wind turbines. The generic nature of the study allows generalization to arbitrary floater designs, based on their natural frequencies in surge and pitch and the associated typical motion amplitudes. While our choice of a non-turbulent inflow allows for a clear investigation of the motion-induced wake mixing, the inclusion of such inflow turbulence is a natural next step. A further extension includes a detailed study of the turbine response for a second turbine within the wake of the present computations. These effects are subject to our present research and will provide further insight into the performance and wake effects in floating wind farms.

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PEER REVIEW

The peer review history for this article is available at <https://publons.com/publon/10.1002/we.2682>.

DATA AVAILABILITY STATEMENT

Data available on request from the authors.

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